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Inventor(s)

Flytzanis; Nicholas C. et al.

SELECTED AAV COMPOSITIONS HAVING PREFERRED BRAIN ENRICHMENT

Abstract

Described herein are compositions and kits comprising recombinant adeno-associated viruses (rAAVs) with increased viral transduction enrichment in the CNS. The rAAV compositions described herein encapsidate a transgene, such as a therapeutic nucleic acid. Gene therapy using the rAAVs is described. Also described are methods of treating CNS-related diseases and conditions.

Inventors: Flytzanis; Nicholas C. (Thousand Oaks, CA), Goeden; Nicholas S. (Thousand Oaks, CA), Sandberg; Troy E. (Thousand Oaks, CA), Wheeler; Brandon G. (Thousand Oaks, CA)

Applicant: Capsida, Inc. (Thousand Oaks, CA)

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Background/Summary

BACKGROUND

[0001] Recombinant adeno-associated viruses (rAAVs) are widely used as vectors for gene delivery in therapeutic applications because of their ability to transduce both dividing and non-dividing cells, their long-term persistence as episomal DNA in infected cells, and their low immunogenicity. These characteristics make them appealing for applications in therapeutic applications, such as gene therapy. However, there is a need to significantly improve the performance of existing AAV serotypes to selectively and efficiently express in distinct cell-types, upon systemic delivery to a subject. This need is especially acute when the AAV must be expressed in the central nervous system (CNS).

SUMMARY OF THE INVENTION

[0002] Disclosed herein are rAAVs with engineered transduction enrichment into the capsid structure through iterative rounds of selection in non-human primates (NHPs), yielding variants having increased transduction enrichment when measured in the CNS.

[0003] The present invention provides rAAVs with widespread transduction to the CNS.

[0004] The present invention provides, in an aspect, a peptide insertion sequence comprising or consisting of an amino-acid sequence set forth in any one of Table 1 and FIG. 1.

[0005] Another aspect of the invention is a modified capsid protein wherein the AAV capsid protein, with a peptide insertion comprising or consisting of an amino-acid sequence set forth in any one of Table 1 and FIG. 1, is characterized by enriched CNS transduction in a subject.

[0006] The present disclosure moreover includes pharmaceutical compositions comprising rAAVs with a peptide insertion comprising or consisting of an amino-acid sequence set forth in any one of Table 1 and FIG. 1, and a pharmaceutically acceptable excipient.

[0007] Aspects disclosed herein provide methods of treating a disease or condition in a subject comprising administering a therapeutically effective amount of a pharmaceutical formulation comprising the AAV capsid protein or the AAV capsid of the present disclosure. In some embodiments, the disease or the condition is a disease or a condition of the CNS, and brain of the subject. Relatedly, the invention includes use of the rAAVs in the manufacture of a medicament for treating or preventing the disease or medical condition.

[0008] Other aspects of the invention will be apparent from the detailed description and claims that follow.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0009] The novel features of the invention are set forth with particularity in the appended claims. A better understanding of the features and advantages of the present invention will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the invention are utilized, and the accompanying drawings of which:

[0010] FIG. 1 shows AAV capsid protein insertion and substitution amino acid sequences and DNA sequences encoding the amino acid sequences which were found in the non-human primate CNS after two rounds of evaluation of engineered AAV libraries.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0011] In one aspect the disclosure provides rAAVs with high expression levels in the CNS.

[0012] In one aspect, the disclosure provides rAAVs with a peptide insertion and or substitution comprising or consisting of an amino-acid sequence set forth in any one of Table 1 and FIG. 1.

[0013] In some embodiments, the amino acid sequence as described in Table 1, is selected from TABLE-US-00001 (SEQ ID NO: 2) AQANTVRNTAQ, (SEQ ID NO: 3) AQDNHVRTVAQ, (SEQ ID NO: 4) AQENSTKSMAQ, (SEQ ID NO: 5) AQENTVRNMAQ, (SEQ ID NO: 6) AQGNNAIRNGAQ, (SEQ ID NO: 7) AQGNNAVKPFAQ, (SEQ ID NO: 8) AQGNNGTRPTAQ, (SEQ ID NO: 9) AQHNNVRPVAQ, (SEQ ID NO: 10) AQLLRDKSSAQ, (SEQ ID NO: 11) AQMNNTAVLAQ, (SEQ ID NO: 12) AQNNTTKLAAQ, (SEQ ID NO: 13) AQNNTVKNLAQ, (SEQ ID NO: 14) AQSNVIRNSAQ, (SEQ ID NO: 15) AQTLKSPATAQ, (SEQ ID NO: 16) AQTNSVKLIAQ, (SEQ ID NO: 15) AQTSVVPVGAQ, (SEQ ID NO: 18) AQYNSTKNMAQ, (SEQ ID NO: 19) AQANSVKNIAQ, and (SEQ ID NO: 20) AQLNNTTKPIDG.

[0014] In some embodiments, the insertion sequence as described in Table 1, is selected from TABLE-US-00002 (SEQ ID NO: 21) ANTVRNT, (SEQ ID NO: 22) DNHVRTV, (SEQ ID NO: 23) ENSTKSM, (SEQ ID NO: 24) ENTVRNM, (SEQ ID NO: 25) GNAIRNG, (SEQ ID NO: 26) GNAVKPF, (SEQ ID NO: 27) GNGTRPT, (SEQ ID NO: 28) HNNVRPV, (SEQ ID NO: 29) LLRDKSS, (SEQ ID NO: 30) MNNTAVL, (SEQ ID NO: 31) NNTTKLA, (SEQ ID NO: 32) NNTVKNL, (SEQ ID NO: 33) SNVIRNS, (SEQ ID NO: 34) TLKSPA, (SEQ ID NO: 35) TNSVKLI, (SEQ ID NO: 36) TSVVPVG, (SEQ ID NO: 37) YNSTKNM, and (SEQ ID NO: 38) ANSVKNI.

[0015] In some embodiments, the insertion sequence is represented by the peptide sequences listed in Table 1.

TABLE-US-00003

TABLE	1	Sequence	SEQ ID NO	Sequence	SEQ ID NO
1	ANTVRNT	21	AQANTVRNTAQ	2	DNHVRTV
2	DNHVRTV	22	AQDNHVRTVAQ	3	ENSTKSM
3	ENSTKSM	23	AQENSTKSMAQ	4	ENTVRNM
4	ENTVRNM	24	AQENTVRNMAQ	5	GNAIRNG
5	GNAIRNG	25	AQGNNAIRNGAQ	6	GNAVKPF
6	GNAVKPF	26	AQGNNAVKPFAQ	7	GNGTRPT
7	GNGTRPT	27	AQGNNGTRPTAQ	8	HNNVRPV
8	HNNVRPV	28	AQHNNVRPVAQ	9	LLRDKSS
9	LLRDKSS	29	AQLLRDKSSAQ	10	MNNTAVL
10	MNNTAVL	30	AQMNNTAVLAQ	11	NNTTKLA
11	NNTTKLA	31	AQNNTTKLAAQ	12	NNTVKNL
12	NNTVKNL	32	AQNNTVKNLAQ	13	SNVIRNS
13	SNVIRNS	33	AQSNVIRNSAQ	14	TLKSPA
14	TLKSPA	34	AQTLKSPATAQ	15	TNSVKLI
15	TNSVKLI	35	AQTNSVKLIAQ	16	TSVVPVG
16	TSVVPVG	36	AQTSVVPVGAQ	17	YNSTKNM
17	YNSTKNM	37	AQYNSTKNMAQ	18	ANSVKNI
18	ANSVKNI	38	AQANSVKNIAQ	19	LNTTKPI
19	LNTTKPI	39	AQLNNTTKPIDG	20	

[0016] In some aspects, the insertion amino acid sequence is at least 71.4% identical to the amino acid sequence provided in Table 1 and FIG. 1. In some aspects, the insertion amino acid sequence is at least 86.7% identical to the amino acid sequence provided in Table 1 and FIG. 1.

[0017] Also disclosed herein are methods and kits for producing therapeutic recombinant AAV (rAAV) particles, as well as methods and pharmaceutical compositions or formulations comprising the rAAV particles, for the treatment of a disease or condition affecting the CNS.

[0018] Disclosed herein are AAV capsids engineered with increased viral transduction in the CNS. The AAV capsids can encapsidate a viral vector with a heterologous nucleic acid encoding, for example, a therapeutic gene expression product. Transduction of the heterologous nucleic acid in the CNS can be achieved upon systemic delivery to a subject of the AAV capsid of the present disclosure encapsidating a heterologous nucleic acid. The AAV capsids disclosed herein are advantageous for many applications of gene therapy to treat human disease, including, but not limited to, disorders of the central nervous system.

[0019] The recombinant AAV vectors comprising a nucleic acid sequence encoding the AAV capsid proteins of the present disclosure as also provided herein. For example, the viral vectors of the

present disclosure comprise a nucleic acid sequence comprising the AAV viral Cap (Capsid) encoding VP1, VP2, and VP3, at least one of which is modified to produce the AAV capsid proteins of the present disclosure. The recombinant AAV vector provided can be derived from an AAV serotype (e.g., AAV9) or a variant AAV serotype including an insertion of the present invention.

AAV Capsids

[0020] Provided herein are modified adeno-associated (AAV) virus capsid compositions useful for integrating a transgene into a target cell or environment (in a subject when they are administered systemically to the subject).

[0021] An rAAV comprises an AAV capsid that can be engineered to encapsidate a heterologous nucleic acid (e.g., therapeutic nucleic acid, gene editing machinery). The AAV capsid is made up of three AAV capsid protein monomers, VP1, VP2, and VP3. Sixty copies of these three VP proteins interact in a 1:1:10 ratio to form the viral capsid. VP1 covers the whole of VP2 protein in addition to a ~137 amino acid N-terminal region (VPlu), VP2 covers the whole of VP3 in addition to ~65 amino acid N-terminal region (VP1/2 common region). The three capsid proteins share a conserved amino acid sequence of VP3, which in some cases is the region beginning at amino acid position 138 (e.g., AA139-736).

[0022] While not wishing to be bound by theory, it is understood that a parent AAV capsid sequence comprises a VP1 region. In certain embodiments, a parent AAV capsid sequence comprises a VP1, VP2 and/or VP3 region, or any combination thereof. A parent VP1 sequence may be considered synonymous with a parent AAV capsid sequence.

[0023] The AAV VP3 structure contains highly conserved regions that are common to all serotypes, a core eight-stranded β -barrel motif (β B- β I) and a small α -helix (α A). The loop regions inserted between the β -strands consist of the distinctive HI loop between β -strands H and I, the DE loop between β -strands D and E, and nine variable regions (VRs), which form the top of the loops. These VRs, such as the AA588 loop, are found on the capsid surface and can be associated with specific functional roles in the AAV life cycle including receptor binding, transduction and antigenic specificity.

[0024] In some aspects, the rAAV variant of the present invention comprises an AAV capsid protein having a peptide insertion at the residues corresponding to amino acids 588-589 of the AAV9 native sequence of SEQ ID NO: 1.

[0025] The AAV capsids comprise AAV capsid proteins (e.g., VP1, VP2, and VP3), each with an insertion, such as in the 588 loop of a parental AAV capsid protein structure (AAV9 VP1 numbering). The 588 loop contains the site of heparan sulfate binding of AAV2 and is amenable to peptide display. The only known receptors for AAV9 is N-linked terminal galactose and AAV receptor (AAVR), but many indications point toward there being others. Modifications to AAV9 588 loop are shown herein to confer an increased transgene transduction in target in vivo environments.

[0026] The present invention provides, in an aspect, a peptide insertion at the AAV 588 loop comprising or consisting of an amino-acid sequence set forth in any one of Table 1 and FIG. 1.

[0027] Disclosed herein are AAV capsids comprising AAV capsid proteins with an insertion at the 588 loop that confer a higher enrichment for transduction in CNS cell types (e.g., brain endothelial cells, neurons, astrocytes). In particular, the AAV capsid proteins disclosed herein enable rAAV-mediated transduction of a heterologous nucleic acid (e.g., transgene) in the CNS of a subject. The AAV capsids of the present disclosure may be formulated as a pharmaceutical composition. In addition, the AAV capsids can be isolated and purified to be used for a variety of applications.

[0028] In some embodiments, the rAAV capsid of the present disclosure are generated using the methods disclosed herein. In some instances, the rAAV capsid is chimeric. In some instances, the rAAV, or variant AAV protein comprises therein, confer an increase in a localization of the rAAV within the target tissue, as compared to the parental AAV capsid or capsid protein.

AAV Capsid Proteins

[0029] Disclosed herein are recombinant AAV (rAAV) capsids which comprise AAV capsid proteins that are engineered with a modified capsid protein (e.g., VP1, VP2, VP3). In some embodiments, the rAAV capsid proteins of the present disclosure are generated using the methods disclosed herein. In some embodiments, the AAV capsid proteins are used in the methods of delivering a therapeutic nucleic acid (e.g., a transgene) to a subject. In some instances, the rAAV capsid proteins have desired AAV expression rendering them particularly suitable for certain therapeutic applications, e.g., the treatment of a disease or disorder in a subject such as those disclosed herein.

[0030] The rAAV capsid proteins are engineered for optimized expression in the CNS, for example the brain, of a subject upon systemic administration of the rAAV to the subject. The rAAV capsid proteins are engineered to include the insertions provided in Table 1 and FIG. 1. The rAAV capsid proteins including the insertions provided in Table 1 and FIG. 1 are engineered to achieve efficient transduction of an encapsidated transgene. In particular, the rAAV capsid proteins have an increased expression enrichment in the brain of a subject.

[0031] The engineered AAV capsid proteins described herein have, in some cases, an insertion of an amino acid that is heterologous to the parental AAV capsid protein at amino acid positions in the 588 loop. In some embodiments, the amino acid is not endogenous to the parental AAV capsid protein at the amino acid position of the insertion. The amino acid may be a naturally occurring amino acid in the same or equivalent amino acid position as the insertion of the substitution in a different AAV capsid protein.

[0032] Generally, the insertion comprises a five-, six-, or seven-amino acid sequence (5-mer, 6-mer, or 7-mer, respectively) that is inserted or substituted at the 588 loop in a parental AAV capsid protein. Aspects provided herein provide amino acid insertions comprising seven amino acid polymer (7-mer) inserted at AA588-589, and may additionally include a substitution of one or two amino acids at amino acid positions flanking the 7-mer sequence (e.g., AA587-588 and/or AA589-590) to produce an eleven amino acid polymer (11-mer) at the 588 loop of a parental AAV capsid protein. The 7-mers described herein were advantageously generated using polymerase chain reaction (PCR) with degenerate primers, where each of the seven amino acids is encoded by a deoxyribose nucleic acid (DNA) sequence N-N-K. “N” is any of the four DNA nucleotides and K is guanine (G) or thymine (T). This method of generating random 7-mer amino acid sequences enables 1.28 billion possible combinations at the protein level.

[0033] The rAAV capsid proteins of the present disclosure comprise an insertion of an amino acid in an amino acid sequence of an AAV capsid protein. The AAV capsid, from which an engineered AAV capsid protein of the present disclosure is produced, is referred to as a “parental” AAV capsid. The complete genome of AAV-1 is provided in GenBank Accession No. NC_002077; the complete genome of AAV-2 is provided in GenBank Accession No. NC_001401 and Srivastava et al., J. Virol., 45:555-564 (1983); the complete genome of AAV-3 is provided in GenBank Accession No. NC_1829; the complete genome of AAV-4 is provided in GenBank Accession No. NC_001829; the AAV-5 genome is provided in GenBank Accession No. AF085716; the complete genome of AAV-6 is provided in GenBank Accession No. NC_001862; at least portions of AAV-7 and AAV-8 genomes are provided in GenBank Accession Nos. AX753246 and AX753249, respectively; the AAV-9 genome is provided in Gao et al., J. Virol., 78:6381-6388 (2004); the AAV-10 genome is provided in Mol. Ther., 13 (1): 67-76 (2006); the AAV-11 genome is provided in Virology, 330 (2): 375-383 (2004); portions of the AAV-12 genome are provided in Genbank Accession No. DQ813647; portions of the AAV-13 genome are provided in Genbank Accession No. EU285562.

[0034] In some cases, the parental AAV is derived from an AAV with a serotype selected from AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11 and AAV12. The AAV capsid protein that is “derived” from another may be a variant AAV capsid protein. A variant may include, for example, a heterologous amino acid in an amino acid sequence of the AAV capsid protein. The heterologous amino acid may be non-naturally occurring in the AAV capsid

protein. The heterologous amino acid may be naturally occurring in a different AAV capsid protein. In some instances, the parental AAV capsid is described in US Pat Publication 2020/0165576 and U.S. Pat. App. Ser. No. 62/832,826 and PCT/US20/20778; the content of each of which is incorporated herein.

[0035] In some instances, the parental AAV is AAV9. In some instances, the amino acid sequence of the AAV9 capsid protein comprises SEQ ID NO: 1. The amino acid sequence of AAV9 VP1 capsid protein (>tr|Q6JC40|Q6JC40_9VIRU Capsid protein VP1 OS=Adeno-associated virus 9 OX=235455 GN=cap PE=1 SV=1) is provided in SEQ ID NO: 1

(MAADGYLPDWLEDNLSEGIREWALKPGAPQPKANQQHQDNARGLVLPGYKYLGPGLGLDKGEPVNAADAAALEHDKAYDQQLKAGDNPYLKYNHADADEFQERLKEDTSFGG NLGRAVFQAKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIGKSGAQPAKKRL NFGQTGDTEVPDPQPIGEPPAAPSGVGS LTMASGGGAPVADNNEGADGVGSSSGNWH CDSQWL GDR VITTSTRTWALPTYNNHLYKQISNSTSGGSSNDNAYFGYSTPWGYFDFN RFHCHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVKEVTDNNGVKTIANNLSTVQVFT DSDYQLPYVLGSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAVGRSSFYCLEYFPSQML RTGNNFQFSYEFENVPFHSSYAHSQSLDRLMNPLIDQYLYLSKTINGSGQNQQTLKFSV AGPSNMAVQGRNYIPGPSYRQQRVSTTVTQNNNSEFAWPGASSWALNGRNSLMNPGP AMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVMITNEEEIKTTNPVATESYGQVA TNHQSAQAQAQTGWVQNQGILPGMVWQDRDVYLQGPWAKIPHTDGNFHPSPLMGGF GMKHPPPQILIKNTPVPADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKENS KRWNPE IQYTSNYYKSNNVEFAVNTEGVYSEPRPIGTRYLTRNL). In some instances, the parental AAV capsid protein sequence is 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% homologous to SEQ ID NO: 1.

[0036] AAV capsid proteins from native AAV serotypes, such as AAV9, with tropisms including the liver activate the innate immune response, which in some cases causes a severe inflammatory response in a subject, which can lead to multi-organ failure. By improving transduction enrichment of a native AAV serotype for a target in vivo tissue (e.g., brain), the rAAV particles of the present disclosure reduce the immunogenic properties of AAV-mediated transgene delivery and prevent activation of the innate immune response.

[0037] In some instances, the parental AAV capsid protein comprises the entire VP1 region provided in SEQ ID NO: 1 (e.g., amino acids 1-736). In some instances, the parental AAV capsid protein comprises amino acids 217-736 in SEQ ID NO: 1, which is the common region found in VP1, VP2 and VP3 AAV9 capsid proteins. In some instances, the AAV capsid protein comprises amino acids 64-736 in SEQ ID NO: 1, which is the common region found in VP1 and VP2. The parental AAV capsid protein sequence may comprise amino acids selected from 1-736, 10-736, 20-736, 30-736, 40-736, 50-736, 60-736, 70-736, 80-736, 90-736, 100-736, 110-736, 120-736, 130-736, 140-736, 150-736, 160-736, 170-736, 180-736, 190-736, 200-736, 210-736, 220-736, 230-736, 240-736, 250-736, 260-736, 270-736, 280-736, 290-736, 300-736, 310-736, 320-736, 330-736, 340-736, 350-736, 360-736, 370-736, 380-736, 390-736, 400-736, 410-736, 420-736, 430-736, 440-736, and 450-736, from SEQ ID NO: 1. In some aspects, the rAAV variant comprises an AAV capsid protein comprising an amino acid sequence that is at least 98% identical to amino acid 217 to amino acid 736 of SEQ ID NO: 1. In some instances, the amino acid insertion is at a three (3)-fold axis of symmetry of a corresponding parental AAV capsid protein.

[0038] Disclosed herein are insertions of an amino acid sequence in an AAV capsid protein. Where the sequence numbering designation “588-589” is noted for AAV9, for example AAV VP1, the invention also includes insertions in similar locations in the other AAV serotypes. As used herein, “AA588-589” indicates that the insertion of the amino acid (or amino acid sequence) is immediately after an amino acid (AA) at position 588 and immediately before an AA at position 589 within an amino acid sequence of a parental AAV VP capsid protein (VP1 numbering). Amino acids 587-591 include a motif comprising “AQAQA” as set forth in SEQ ID NO: 1. Exemplary

AAV capsid protein sequences are provided in Table 2. For example, ANTVRNT (SEQ ID NO: 21) is inserted at AA588-589 in an AAV9 capsid amino acid sequence, and provides variant A (SEQ ID NO: 45). It is envisioned that the insertions disclosed herein (Table 1 and FIG. 1) may be inserted at AA588-589 in an amino acid sequence of a parental AAV9 capsid protein, a variant thereof, or equivalent amino acid position of a parental AAV of a different serotype (e.g., AAV1, AAV2, AAV3, and the like). In any AAV capsid protein sequence disclosed herein, the amino acid at position 449 may be R or K.

TABLE-US-00004 TABLE 2 Exemplary AAV Capsid Protein Sequences SEQ ID Identifier Sequence 40 AAV-PHP. MAADGYLPDWLEDNLSEGIREWWALKPGAPQP eB

KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADAEFQERLKEDTSFGGNLGRAVFQAKKRLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSSGNWHCDSQWLGDRVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFPRKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVMITNEEEIKTTNPVATESYGQVATNHQS
DGTLAVPFKAQAQTGWVQNQGILPGMVWQDRDVYLQGP
IWAKIPHTDGNFHPSPLMGGFGMKHPPPQILIKNTPVPADPPTAFNKDKLNSFITQYSTGQVSVEIEWELQKENS
KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL

41 AAV.CAP- MAADGYLPDWLEDNLSEGIREWWALKPGAPQP A4

KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADAEFQERLKEDTSFGGNLGRAVFQAKKRLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSSGNWHCDSQWLGDRVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFPRKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRT**IKDNTP**
GRQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQAQAQTGWVQNQGILPGMVWQDRDVYLQGP
IWAKIPHTDGNFHPSPLMGGFGMKHPPPQILIKN

TPVPADPDKLTNSFITQYSTGQVSVEIEW
ELQKENS KRWNPEIQYTSNYYKSNNVEFAVNTE GVVSEPRPIGTRYLTRNL 42 AAV.CAP-
MAADGYLPDWLEDNLSEGIREWWALKPGAPQP B2
KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGD RVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFS PRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTI**QQGKQ**
SVQTLKFSVAGPSNMAVQGRNYIPGPSYRQQRV
STTVTQNNNSEFAWPGASSWALNGRNSLMNPG
PAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVD
ADKVMITNEEEIKTTNPVATESYGQVATNHQSD
GTLAVPFKAQAQTGWVQNQGILPGMVWQDRD
VYLQGP IWAKIPHTDGNFHPSPLMGGFGMKHPP
PQILIKNTPVPADPPTAFNKDKLNSFITQYSTGQ
VSVEIEWELQKENS KRWNPEIQYTSNYYKSNNVE FAVNTEGVVSEPRPIGTRYLTRNL 43
AAV.CAP- MAADGYLPDWLEDNLSEGIREWWALKPGAPQP B10
KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGD RVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFS PRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTI**DGAAT**
KNQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
DGTLAVPFKAQAQTGWVQNQGILPGMVWQDR
DVYLQGP IWAKIPHTDGNFHPSPLMGGFGMKHP
PPQILIKNTPVPADPPTAFNKDKLNSFITQYSTG
QVSVEIEWELQKENS KRWNPEIQYTSNYYKSNNV EFAVNTEGVVSEPRPIGTRYLTRNL 44
AAV.CAP- MAADGYLPDWLEDNLSEGIREWWALKPGAPQP B22
KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL

EPLGLVEEAAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGDREVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYYLSRT**IDGQSS**
KSQTLKFSVAGPSNMAVQGRNYIPGPSYRQQRV
STTVTQNNNSEFAWPGASSWALNGRNSLMNPG
PAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVD
ADKVMITNEEEIKTTNPVATESYGQVATNHQSD
GTLAVPFKAQAQTGWVQNQGILPGMVWQDRD
VYLQGPIWAKIPHTDGNFHPSPLMGGFGMKHPP
PQILIKNTPVPADPPTAFNKDKLNSFITQYSTGQ
VSVEIEWELQKENS KRWNPEIQYTSNYYKSNNVE FAVNTEGVYSEPRPIGTRYLTRNL 45
Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP A
KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGDREVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQANTVRNTAQAQTGWVQNQGILPGMVWQD
RDVYLQGPIWAKIPHTDGNFHPSPLMGGFGMK
HPPPQILIKNTPVPADPPTAFNKDKLNSFITQYS
TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
46 Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP B
KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGDREVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK

EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQDNHVRTVAQAQTGWVQNQGILPGMVWQD
RDVYLQGPIWAKIPHTDGNFHPSPLMGGFGMK
HPPPQILIKNTPVPADPPTAFNKDKLNSFITQYS
TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
47 Variant MAADGYLPDWLEDNLSEGIREW WALKPGAPQP C

KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTE SVDPQP IGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGD RVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFS PRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQENSTKSMAQAQTGWVQNQGILPGMVWQD
RDVYLQGPIWAKIPHTDGNFHPSPLMGGFGMK
HPPPQILIKNTPVPADPPTAFNKDKLNSFITQYS
TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
48 Variant MAADGYLPDWLEDNLSEGIREW WALKPGAPQP D

KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTE SVDPQP IGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGD RVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFS PRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP

GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQENTVRNMAQAQTGWVQNQGILPGMVWQD
RDVYLQGPWIWAKIPHTDGNFHPSPLMGGFGMK
HPPPQILIKNTPVPADPPTAFNKDKLNSFITQYS
TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
49 Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP E

KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTE SVDPDPPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGD RVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFS PRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVF TDS DYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQLNTTKPIDGAQQTGWVQNQGILPGMVWQDR
DVYLQGPWIWAKIPHTDGNFHPSPLMGGFGMKHP
PPQILIKNTPVPADPPTAFNKDKLNSFITQYSTG

QVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NV EFAVNTEGVYSEPRPIGTRYLTRNL 50
Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP F

KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTE SVDPDPPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGD RVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFS PRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVF TDS DYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQGNAIRNGAQAQTGWVQNQGILPGMVWQD
RDVYLQGPWIWAKIPHTDGNFHPSPLMGGFGMK
HPPPQILIKNTPVPADPPTAFNKDKLNSFITQYS
TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL

51 Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP G

KANQQHQDNARGLVLPGYKYLGPNGGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADADEFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGDRVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
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GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQSNVIRNSAQAQTGWVQNGILPGMVWQDR
DVYLGQPIWAKIPHTDGNFHPSPLMGGFGMKHP
PPQILIKNTPVPADPPTAFNKDKLNSFITQYSTG

QVSVEIEWELQKENSKRWNPEIQYTSNYYKSNNV EFAVNTEGVYSEPRPIGTRYLTRNL 52

Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP H

KANQQHQDNARGLVLPGYKYLGPNGGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADADEFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGDRVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQANSVKNIAQAQTGWVQNGILPGMVWQDR
DVYLGQPIWAKIPHTDGNFHPSPLMGGFGMKHP
PPQILIKNTPVPADPPTAFNKDKLNSFITQYSTG

QVSVEIEWELQKENSKRWNPEIQYTSNYYKSNNV EFAVNTEGVYSEPRPIGTRYLTRNL 53

Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP I

KANQQHQDNARGLVLPGYKYLGPNGGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADADEFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA

APSGVGSALTMASSGGAPVADNNEGADGVGSSS
GNWHCDSQWLGDREVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGVATNHQS
AQYNSTKNMAQAQTGWVQNNQGILPGMVWQD
RDVYLQGPWAKIPHTDGNFHPSPLMGGFGMK
HPPPQILIKNTPVPADPPTAFNKDKLNSFITQYS
TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
54 Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP J

KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADADEFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSALTMASSGGAPVADNNEGADGVGSSS
GNWHCDSQWLGDREVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGVATNHQS
AQTLKSPATAQAQTGWVQNNQGILPGMVWQDR
DVYLQGPWAKIPHTDGNFHPSPLMGGFGMKHP
PPQILIKNTPVPADPPTAFNKDKLNSFITQYSTG
QVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NV EFAVNTEGVYSEPRPIGTRYLTRNL 55
Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP K

KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADADEFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSALTMASSGGAPVADNNEGADGVGSSS
GNWHCDSQWLGDREVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV

GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQGNVAKPFAQAQTGWVQNQGILPGMVWQD
RDVYLQGPIWAKIPHTDGNFHPSPLMGGFGMK
HPPPQILIKNTPVPADPPTAFNKDKLNSFITQYS
TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
58 Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP L

KANQQHQDNARGLVLPGYKYLPGNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGDRVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFS PRDWQRLINNNWGF RPKRLNFKLFNIQVK
EVT DNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
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AQGNGRPTAQAQTGWVQNQGILPGMVWQD
RDVYLQGPIWAKIPHTDGNFHPSPLMGGFGMK
HPPPQILIKNTPVPADPPTAFNKDKLNSFITQYS
TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
59 Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP M

KANQQHQDNARGLVLPGYKYLPGNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGDRVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFS PRDWQRLINNNWGF RPKRLNFKLFNIQVK
EVT DNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS

AQHNNVRPQAQTGWVQNQGILPGMVWQD
RDVYLQGPIWAKIPHTDGNFHPSPLMGGFGMK
HPPPQILIKNTPVPADPPTAFNKDKLNSFITQYS
TGQVSVEIEWELQKENSKRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
60 Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP N
KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADADEFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGDREVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQLLRDKSSAQATGWVQNQGILPGMVWQDR
DVYLQGPIWAKIPHTDGNFHPSPLMGGFGMKHP
PPQILIKNTPVPADPPTAFNKDKLNSFITQYSTG
QVSVEIEWELQKENSKRWNPEIQYTSNYYKSNNV EFAVNTEGVYSEPRPIGTRYLTRNL 61
Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP O
KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADADEFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTEVPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGDREVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
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AQMNNTAVLAQATGWVQNQGILPGMVWQD
RDVYLQGPIWAKIPHTDGNFHPSPLMGGFGMK
HPPPQILIKNTPVPADPPTAFNKDKLNSFITQYS
TGQVSVEIEWELQKENSKRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL
62 Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP P
KANQQHQDNARGLVLPGYKYLGPNGLDKGE

PVNAADAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTE SVDPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGD RVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
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GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQNNTTKLAAQAQTGWVQNQGILPGMVWQD
RDVYLQGPIWAKIPHTDGNFHPSPLMGGFGMK
THPPPQILIKNTPVPADPPTAFNKDKLNSFITQYS

GQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN 63 Variant

NVEFAVNTEGVYSEPRPIGTRYLTRNL Q
MAADGYLPDWLEDNLSEGIREWWALKPGAPQP
KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTE SVDPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGD RVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTPWGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS

AQNNTVKNLAQAQTGWVQNQGILPGMVWQD
RDVYLQGPIWAKIPHTDGNFHPSPLMGGFGMK
HPPPQILIKNTPVPADPPTAFNKDKLNSFITQYS

TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL

64 Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP R

KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTE SVDPDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS

GNWHCDSQWLGD RVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTP WGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQSNVIRNSAQAQTGWVQNQGILPGMVWQDR
DVYLGQPIWAKIPHTDGNFHPSPLMGGFGMKHP
PPQILIKNTPVPADPPTAFNKDKLNSFITQYSTG
QVSVEIEWELQKENS KRWNPEIQYTSNYYKSNNV EFAVNTEGVYSEPRPIGTRYLTRNL 65
Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP S

KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTE SVDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGD RVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTP WGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH
SSYAHSQSLDRLMNPLIDQYLYLSRTINGSGQ
NQQTLKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQTNSVKLIAQAQTGWVQNQGILPGMVWQDR
DVYLGQPIWAKIPHTDGNFHPSPLMGGFGMKHP
PPQILIKNTPVPADPPTAFNKDKLNSFITQYSTG
QVSVEIEWELQKENS KRWNPEIQYTSNYYKSNNV EFAVNTEGVYSEPRPIGTRYLTRNL 66
Variant MAADGYLPDWLEDNLSEGIREWWALKPGAPQP T

KANQQHQDNARGLVLPGYKYLGPNGLDKGE
PVNAADAAALEHDKAYDQQLKAGDNPYLKYN
HADA EFQERLKEDTSFGGNLGRAVFQAKKRLL
EPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIG
KSGAQPAKKRLNFGQTGDTE SVDPQPIGEPPA
APSGVGSLTMASGGGAPVADNNEGADGVGSSS
GNWHCDSQWLGD RVITTSTRTWALPTYNNHLY
KQISNSTSGGSSNDNAYFGYSTP WGYFDFNRFH
CHFSPRDWQRLINNNWGFRPKRLNFKLFNIQVK
EVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVL
GSAHEGCLPPFPADVFMIPQYGYLTLNDGSQAV
GRSSFYCLEYFPSQMLRTGNNFQFSYEFENVPFH

SSYAHQSGLDRLMNLIDQYLYLSRTINGSGQ
NQQLTKFSVAGPSNMAVQGRNYIPGPSYRQQR
VSTTVTQNNNSEFAWPGASSWALNGRNSLMNP
GPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNV
DADKVMITNEEEIKTTNPVATESYGQVATNHQS
AQTSVVPVGAQAAQTGWVQNQGILPGMVWQD
RDVYLQGPWIWAKIPHTDGNFHPSPLMGGFGMK
HPPPQILIKNTPVPADPPTAFNKDKLNSFITQYS
TGQVSVEIEWELQKENS KRWNPEIQYTSNYYKSN NVEFAVNTEGVYSEPRPIGTRYLTRNL

[0039] The insertions described herein may, in some cases, comprise a 7-mer insertion at AA588-589. It is envisioned that any 7-mer insertion disclosed herein in addition to a substitution with any amino acid at amino acid positions 587-590 [AQAQ] may comprise an 11-mer.

[0040] Disclosed herein are AAV capsid proteins with an insertion described above in a parental AAV capsid protein that confers an increased transduction enrichment in the CNS in a subject, even when delivered systemically. One of the many advantages of the AAV capsid proteins described herein is their ability to target tissue and cells within the CNS. The tissue can be the brain. Non-limiting examples of CNS cells include a neuron and a glial cell. Glial cells can be selected from an oligodendrocyte, an ependymal cell, an astrocyte and a microglia.

[0041] In some instances, the AAV capsid protein comprises an insertion of at least or about five, six, or seven amino acids of an amino acid sequence of Table 1 and FIG. 1 at an amino acid position 588-589 in a parental AAV9 capsid protein (SEQ ID NO: 1). In some cases, the AAV capsid protein has increased viral transduction in brain.

[0042] The rAAV capsid proteins of the present disclosure may also have a substitution of an amino acid sequence at amino acid position 452-458 in a parental AAV9 capsid protein, or variant thereof, as described in WO2020068990. In some embodiments, the substitution of the amino acid sequence comprises KDNTTPGR (SEQ ID NO: 367) at amino acid position 452-458 in the parental AAV9 capsid protein. In some embodiments, the substitution of the amino acid sequence comprises DGAATKN (SEQ ID NO: 368) at amino acid position 452-458 in the parental AAV9 capsid protein.

[0043] The rAAV capsid proteins described herein may be isolated and purified. The AAV may be isolated and purified by methods standard in the art such as by column chromatography, iodixanol gradients, or cesium chloride gradients. Methods for purifying AAV from helper virus are known in the art and may include methods disclosed in, for example, Clark et al., Hum. Gene Ther., 10 (6): 1031-1039 (1999); Schenpp and Clark, Methods Mol. Med., 69:427-443 (2002); U.S. Pat. No. 6,566,118 and WO 98/09657.

[0044] In addition, the AAV capsid proteins disclosed herein, either isolated and purified, or not, may be formulated into a pharmaceutical formulation, which in some cases, further comprises a pharmaceutically acceptable carrier.

[0045] The rAAV capsid protein can be conjugated to a nanoparticle, a second molecule, or a viral capsid protein. In some cases, the nanoparticle or viral capsid protein would encapsidate the therapeutic nucleic acid described herein. In some instances, the second molecule is a therapeutic agent, e.g., a small molecule, antibody, antigen-binding fragment, peptide, or protein, such as those described herein.

[0046] "Percent Identity" is the percent of the symbols that actually match. Percent Similarity is the percent of the symbols that are similar. Symbols that are across from gaps are ignored. A similarity is scored when the scoring matrix value for a pair of symbols is greater than or equal to 0.50, the similarity threshold. The scoring matrix used in Version 10 of the Wisconsin Genetics Software Package is BLOSUM62 (see: Henikoff and Henikoff, (1989) Proc. Natl. Acad. Sci. USA 89:10915).

[0047] Sequence identity/similarity values provided herein can refer to the value obtained using the

BLAST+2.5.0 suite of programs using default settings (blast.ncbi.nlm.nih.gov) (Camacho, C., et al. (2009) BLAST+: architecture and applications. BMC Bioinformatics 10:421).

[0048] As those of ordinary skill in the art will understand, BLAST searches assume that proteins can be modeled as random sequences. However, many real proteins comprise regions of nonrandom sequences, which may be homopolymeric tracts, short-period repeats, or regions enriched in one or more amino acids. Such low-complexity regions may be aligned between unrelated proteins even though other regions of the protein are entirely dissimilar. A number of low-complexity filter programs can be employed to reduce such low-complexity alignments. For example, the SEG (Wooten and Federhen, (1993) Comput. Chem. 17:149-63) and XNU (Ci-ayerie and States (1993) Comput. Chem. 17:191-201) low-complexity filters can be employed alone or in combination.

[0049] The terms “substantial identity” and “substantially identical” indicate that a polypeptide or nucleic acid comprises a sequence with between 55-100% sequence identity to a reference sequence, with at least 55% sequence identity, or at least 60%, or at least 65%, or at least 70%, or at least 75%, or at least 80%, or at least 85%, or at least 90%, or at least 95%, or at least 99% sequence identity or any percentage of value within the range of 55-100% sequence identity relative to the reference sequence. The percent sequence identity may occur over a specified comparison window. Optimal alignment may be ascertained or conducted using the homology alignment algorithm of Needleman and Wunsch, supra.

[0050] For example, the insertion sequences may include, but are not limited to, sequences that are not exactly the same as the sequences disclosed herein, but which have, in addition to the substitutions explicitly described for various sequences listed herein, additional substitutions of amino acid residues which substantially do not impair the activity or properties of the sequences described herein, such as those predicted by homology software e.g. BLOSUM62 matrices.

AAV Particles

[0051] The rAAV particles with the insertion sequences described herein have an increased transduction enrichment in the CNS. In some instances, the increased transduction enrichment comprises a 1-fold, 2-fold, 3-fold, 4-fold, 5-fold, 6-fold, 7-fold, 8-fold, 9-fold or 10-fold increase, or more. In some instances, the increased transduction enrichment is at least 1-fold. In some instances, the increased transduction enrichment is at least 2-fold. In some instances, the increased transduction enrichment is at least 4-fold.

[0052] The rAAV particles with the insertion sequences described herein have an increased expression enrichment in the CNS. Detecting whether a rAAV possesses more or less expression includes measuring a level of gene expression product (e.g., RNA or protein) expressed from the heterologous nucleic acid encapsidated by the rAAV in a tissue sample obtained from a subject. Suitable methods for measuring expression of a gene expression product include next-generation sequencing (NGS) and quantitative polymerase chain reaction (qPCR).

[0053] The increased expression in the CNS is represented by the enrichment values provided in FIG. 1.

Heterologous Nucleic Acids

[0054] Disclosed herein are therapeutic nucleic acids useful for the treatment or prevention of a disease or condition, or symptom of the disease or condition. In some embodiments, the therapeutic nucleic acids encode a therapeutic gene expression product. Non-limiting examples of gene expression products include proteins, polypeptides, peptides, enzymes, antibodies, antigen binding fragments, nucleic acid (RNA, DNA, antisense oligonucleotide, siRNA, and the like), and gene editing components, for use in the treatment, prophylaxis, and/or amelioration of the disease or disorder, or symptoms of the disease or disorder. In some instances, the therapeutic nucleic acids are placed in an organism, cell, tissue or organ of a subject by way of a rAAV, such as those disclosed herein.

[0055] Disclosed herein are rAAVs, each comprising a viral vector (e.g., a single stranded DNA

molecule (ssDNA)). In some instances, the viral vector comprises two inverted terminal repeat (ITR) sequences that are about 145 bases each, flanking a transgene. In some embodiments, the transgene comprises a therapeutic nucleic acid, and in some cases, a promoter in cis with the therapeutic nucleic acid in an open reading frame (ORF). The promoter is capable of initiating transcription of therapeutic nucleic acid in the nucleus of the target cell. The ITR sequences can be from any AAV serotype. Non-limiting examples of AAV serotypes include AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, and AAV12. In some cases, an ITR is from AAV2. In some cases, an ITR is from AAV9.

[0056] Disclosed herein are transgenes that can comprise any number of nucleotides. In some cases, a transgene can comprise less than about 100 nucleotides. In some cases, a transgene can comprise at least about 100 nucleotides. In some cases, a transgene can comprise at least about 200 nucleotides. In some cases, a transgene can comprise at least about 300 nucleotides. In some cases, a transgene can comprise at least about 400 nucleotides. In some cases, a transgene can comprise at least about 500 nucleotides. In some cases, a transgene can comprise at least about 1000 nucleotides. In some cases, a transgene can comprise at least about 5000 nucleotides. In some cases, a transgene can comprise over 5,000 nucleotides. In some cases, a transgene can comprise between about 500 and about 5000 nucleotides. In some cases, a transgene comprises about 5000 nucleotides. In any of the cases disclosed herein, the transgene can comprise DNA, RNA, or a hybrid of DNA and RNA. In some cases, the transgene can be single stranded. In some cases, the transgene can be double stranded.

[0057] Disclosed herein are transgenes useful for modulating the expression or activity of a target gene or gene expression product thereof. In some instances, the transgene is encapsidated by an rAAV capsid protein of an rAAV particle described herein. In some instances, the rAAV particle is delivered to a subject to treat a disease or condition disclosed herein in the subject. In some instances, the delivery is systemic.

[0058] The transgenes disclosed herein are useful for expressing an endogenous gene at a level similar to that of a healthy or normal individual. This is particularly useful in the treatment of a disease or condition related to the underexpression, or lack of expression, of a gene expression product. In some embodiments, the transgenes disclosed herein are useful for overexpressing an endogenous gene, such that an expression level of the endogenous gene is above the expression level of a healthy or normal individual. Additionally, transgenes can be used to express exogenous genes (e.g., active agent such as an antibody, peptide, nucleic acid, or gene editing components). In some embodiments, the therapeutic gene expression product is capable of altering, enhancing, increasing, or inducing the activity of one or more endogenous biological processes in the cell. In some embodiments, the transgenes disclosed herein are useful for reducing expression of an endogenous gene, for example, a dominant negative gene. In some embodiments, the therapeutic gene expression product is capable of altering, inhibiting, reducing, preventing, eliminating, or impairing the activity of one or more endogenous biological processes in the cell. In some aspects, the increase of gene expression refers to an increase by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95% and 100%. In one aspect, the protein product of the targeted gene may be increased by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95% and 100%. In some aspects, the decrease of gene expression refers to a decrease by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95% and 100%. In one aspect, the protein product of the targeted gene may be decreased by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95% and 100%.

[0059] When endogenous sequences (endogenous or part of a transgene) are expressed with a transgene, the endogenous sequences can be full-length sequences (wild-type or mutant) or partial sequences. The endogenous sequences can be functional. Non-limiting examples of the function of these full length or partial sequences include increasing the serum half-life of the polypeptide expressed by a transgene (e.g., therapeutic gene) and/or acting as a carrier.

[0060] A transgene can be inserted into an endogenous gene such that all, some or none of the endogenous gene is expressed. For example, a transgene as described herein can be inserted into an endogenous locus such that some (N-terminal and/or C-terminal to a transgene) or none of the endogenous sequences are expressed, for example as a fusion with a transgene. In other cases, a transgene (e.g., with or without additional coding sequences of the endogenous gene) is integrated into any endogenous locus, for example a safe-harbor locus. For example, a Frataxin (FXN) transgene can be inserted into an endogenous FXN gene. A transgene can be inserted into any gene, e.g., the genes as described herein.

[0061] At least one advantage of the present disclosure is that virtually any therapeutic nucleic acid may be used to express any therapeutic gene expression product. In some instances, the therapeutic gene expression product is a therapeutic protein or a peptide (e.g., antibody, antigen-binding fragment, peptide, or protein). In one embodiment the protein encoded by the therapeutic nucleic acid is between 50-5000 amino acids in length. In some embodiments the protein encoded is between 50-2000 amino acids in length. In some embodiments the protein encoded is between 50-1000 amino acids in length. In some embodiments the protein encoded is between 50-1500 amino acids in length. In some embodiments the protein encoded is between 50-800 amino acids in length. In some embodiments the protein encoded is between 50-600 amino acids in length. In some embodiments the protein encoded is between 50-400 amino acids in length. In some embodiments the protein encoded is between 50-200 amino acids in length. In some embodiments the protein encoded is between 50-100 amino acids in length. In some embodiments the peptide encoded is between 4-50 amino acids in length. In some embodiments, the protein encoded is a tetrapeptide, a pentapeptide, a hexapeptide, a heptapeptide, an octapeptide, a nonapeptide, or a decapeptide. In some embodiments, the protein encoded comprises a peptide of 2-30 amino acids, such as for example 5-30, 10-30, 2-25, 5-25, 10-25, or 10-20 amino acids. In some embodiments, the protein encoded comprises a peptide of at least 11, 12, 13, 14, 15, 17, 20, 25 or 30 amino acids, or a peptide that is no longer than 50 amino acids, e.g., no longer than 35, 30, 25, 20, 17, 15, 14, 13, 12, 11 or 10 amino acids.

[0062] Non-limiting examples of therapeutic protein or peptides include an adrenergic agonist, an anti-apoptosis factor, an apoptosis inhibitor, a cytokine receptor, a cytokine, a cytotoxin, an erythropoietic agent, a glutamic acid decarboxylase, a glycoprotein, a growth factor, a growth factor receptor, a hormone, a hormone receptor, an interferon, an interleukin, an interleukin receptor, a kinase, a kinase inhibitor, a nerve growth factor, a netrin, a neuroactive peptide, a neuroactive peptide receptor, a neurogenic factor, a neurogenic factor receptor, a neuropilin, a neurotrophic factor, a neurotrophin, a neurotrophin receptor, an N-methyl-D-aspartate antagonist, a plexin, a protease, a protease inhibitor, a protein decarboxylase, a protein kinase, a protein kinase inhibitor, a proteolytic protein, a proteolytic protein inhibitor, a semaphoring, a semaphorin receptor, a serotonin transport protein, a serotonin uptake inhibitor, a serotonin receptor, a serpin, a serpin receptor, and a tumor suppressor. In certain embodiments, the therapeutic protein or peptide is selected from brain-derived neurotrophic factor (BDNF), ciliary neurotrophic factor (CNTF), macrophage colony-stimulating factor (CSF), epidermal growth factor (EGF), fibroblast growth factor (FGF), gonadotropin, interferon-gamma (IFN), insulin-like growth factor 1 (IGF-1), nerve growth factor (NGF), platelet-derived growth factor (PDGF), pigment epithelium-derived factor (PEDF), transforming growth factor (TGF), transforming growth factor-beta (TGF-B), tumor necrosis factor (TNF), vascular endothelial growth factor (VEGF), prolactin, somatotropin, X-linked inhibitor of apoptosis protein 1 (XIAP1), interleukin 1 (IL-1), IL-2, IL-3, IL-4, IL-5, IL-6, IL-7, IL-8, IL-9, IL-10, IL-10, viral IL-10, IL-11, IL-12, IL-13, IL-14, IL-15, IL-16, IL-17, and IL-18.

[0063] A therapeutic gene expression product can comprise gene editing components. Non-limiting examples of gene editing components include those required for CRISPR/Cas, artificial site-specific RNA endonuclease (ASRE), zinc finger endonuclease (ZFN), and transcription factor like

effector nuclease (TALEN). In a non-limiting example, a subject having Huntington's disease is identified. The subject is then systemically administered a first amount of a rAAV encapsidating a viral vector encoding ZFN engineered to represses the transcription of the Huntingtin (HTT) gene. The rAAV will include a modified AAV capsid protein that includes an amino acid sequence provided in any one of Table 1 and FIG. 1, so as to allow proper targeting of the ZFN to the nervous system, while reducing expression in off-target organs, such as the liver. If needed, the subject is administered a second or third dose of the rAAV, until a therapeutically effective amount of the ZEN is expressed in the subject's nervous system.

[0064] A therapeutic nucleic acid can comprise a non-protein coding gene e.g., sequences encoding antisense RNAs, RNAi, shRNAs and micro RNAs (miRNAs), miRNA sponges or decoys, recombinase delivery for conditional gene deletion, conditional (recombinase-dependent) expression, includes those required for the gene editing components described herein. The non-protein coding gene may also encode a tRNA, IRNA, tmRNA, piRNA, double stranded RNA, snRNA, snoRNA, and/or long non-coding RNA (lncRNA). In some cases, the non-protein coding gene can modulate the expression or the activity of a target gene or gene expression product. For example, the RNAs described herein may be used to inhibit gene expression in the CNS. In some cases, inhibition of gene expression refers to an inhibition by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95% and 100%. In some cases, the protein product of the targeted gene may be inhibited by at least about 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95% and 100%. The gene can be either a wild type gene or a gene with at least one mutation. The targeted protein may be either a wild type protein or a protein with at least one mutation.

[0065] A therapeutic nucleic acid can modulate the expression or activity of a gene or gene expression product expressed from the gene that is implicated in a disease or disorder of the CNS. For example, the therapeutic nucleic acid, in some cases is a gene or a modified version of the gene described herein. In some instances, the gene or gene expression product is inhibited. In some instances, the gene or gene expression product is enhanced.

[0066] In another example, the therapeutic nucleic acid comprises an effector gene expression product such as a gene editing component specific to target a gene therein. Non-limited examples of genes include target gene or gene expression product selected from ATP1A2, CACNA1A, SETD5, SHANK3, NF2, DNMT1, TCF4, RAI1, PEX1, ARSA, EIF2B5, EIF2B1, EIF2B2, NPC1, ADAR, MFSD8, STXBPI, PRICKLE2, PRRT2, IDUA, STX1B, Sarcoglycan Alpha (SGCA), glutamic acid decarboxylase 65 (GAD65), glutamic acid decarboxylase 67 (GAD67), CLN2, Nerve Growth Factor (NGF), glial cell derived neurotrophic factor (GDNF), Survival Of Motor Neuron 1, STXBP1, Telomeric (SMNI), Factor X (FIX), Retinoid Isomerohydrolase (RPE65), sarco/endoplasmic reticulum Ca²⁺-ATPase (SERCA2a), Glucocerebrosidase (GCase), galactocerebrosidase (GALC), CDKL5, Frataxin (FXN), Huntingtin (HTT), methyl-CpG binding protein 2 (MECP2), a peroxisomal biogenesis factor (PEX), progranulin (GRN), an antitubulin agent, copper-zinc superoxide dismutase (SOD1), iduronate 2 sulfatase (hIDS), Glucosylceramidase Beta (GBA), fragile X mental retardation 1 (FMR1), NPC Intracellular Cholesterol Transporter 1 (NPC1), SCN1A, C9orf72, NPS3 and a NLRP3 inflammasome. In some embodiments, the peroxisomal biogenesis factor (PEX) is selected from PEX1, PEX2, PEX3, PEX4, PEX5, PEX6, PEX7, PEX10, PEX11 β , PEX12, PEX13, PEX14, PEX16, PEX19, and PEX26. In some instances, the gene or gene expression product is inhibited. In some instances, the gene or gene expression product is enhanced.

AAV Vectors

[0067] Aspects disclosed herein comprise plasmid vectors comprising a nucleic acid sequence encoding the AAV capsids and AAV capsid proteins described herein. AAV vectors described herein are useful for the assembly of a rAAV and viral packaging of a heterologous nucleic acid. In addition, an AAV vector may encode a transgene comprising the heterologous nucleic acid.

[0068] An AAV vector can comprise a transgene, which in some cases encodes a heterologous gene

expression product (e.g., therapeutic gene expression product, recombinant capsid protein, and the like). The transgene is in cis with two inverted terminal repeats (ITRs) flanking the transgene. The transgene may comprise a therapeutic nucleic acid encoding a therapeutic gene expression product. Due to the limited packaging capacity of the rAAV (~5 kB), in some cases, a longer transgene may be split between two AAV vectors, the first with 3' splice donor and the second with a 5' splice acceptor. Upon co-infection of a cell, concatemers form, which are spliced together to express a full-length transgene.

[0069] A transgene is generally inserted so that its expression is driven by the endogenous promoter at the integration site, namely the promoter that drives expression of the endogenous gene into which a transgene is inserted. In some instances, a transgene comprises a promoter and/or enhancer, for example a constitutive promoter or an inducible or tissue/cell specific promoter. As a non-limiting example, the promoter may be CMV promoter, a CMV- β -Actin-intron- β -Globin hybrid promoter (CAG), CBA promoter, FRDA or FXN promoter, UBC promoter, GUSB promoter, NSE promoter, Synapsin promoter, MeCP2 promoter, GFAP promoter, H1 promoter, U6 promoter, NFL promoter, NFH promoter, SCN8A promoter, or PGK promoter. As a non-limiting example, promoters can be tissue-specific expression elements include, but are not limited to, human elongation factor 1 α -subunit (EF1 α), immediate-early cytomegalovirus (CMV), chicken β -actin (CBA) and its derivative CAG, the β glucuronidase (GUSB), and ubiquitin C (UBC). The transgene may include a tissue-specific expression elements for neurons such as, but not limited to, neuron-specific enolase (NSE), platelet-derived growth factor (PDGF), platelet-derived growth factor B-chain (PDGF- β), the synapsin (Syn), the methyl-CpG binding protein 2 (MeCP2), Ca²⁺/calmodulin-dependent protein kinase II (CaMKII), metabotropic glutamate receptor 2 (mGluR2), NFL, NFH, np32, PPE, Enk and EAAT2 promoters. The transgene may comprise a tissue-specific expression element for astrocytes such as, but not limited to, the glial fibrillary acidic protein (GFAP) and EAAT2 promoters. The transgene may comprise tissue-specific expression elements for oligodendrocytes such as, but not limited to, the myelin basic protein (MBP) promoter.

[0070] In some embodiments, the promoter is less than 1 kb. The promoter may have a length of 200, 210, 220, 230, 240, 250, 260, 270, 280, 290, 300, 310, 320, 330, 340, 350, 360, 370, 380, 390, 400, 410, 420, 430, 440, 450, 460, 470, 480, 490, 500, 510, 520, 530, 540, 550, 560, 570, 580, 590, 600, 610, 620, 630, 640, 650, 660, 670, 680, 690, 700, 710, 720, 730, 740, 750, 760, 770, 780, 790, 800 or more than 800. The promoter may have a length between 200-300, 200-400, 200-500, 200-600, 200-700, 200-800, 300-400, 300-500, 300-600, 300-700, 300-800, 400-500, 400-600, 400-700, 400-800, 500-600, 500-700, 500-800, 600-700, 600-800 or 700-800. The promoter may provide expression of the therapeutic gene expression product for a period of time in targeted tissues such as, but not limited to, the CNS. Expression of the therapeutic gene expression product may be for a period of 1 hour, 2, hours, 3 hours, 4 hours, 5 hours, 6 hours, 7 hours, 8 hours, 9 hours, 10 hours, 11 hours, 12 hours, 13 hours, 14 hours, 15 hours, 16 hours, 17 hours, 18 hours, 19 hours, 20 hours, 21 hours, 22 hours, 23 hours, 1 day, 2 days, 3 days, 4 days, 5 days, 6 days, 1 week, 8 days, 9 days, 10 days, 11 days, 12 days, 13 days, 2 weeks, 15 days, 16 days, 17 days, 18 days, 19 days, 20 days, 3 weeks, 22 days, 23 days, 24 days, 25 days, 26 days, 27 days, 28 days, 29 days, 30 days, 31 days, 1 month, 2 months, 3 months, 4 months, 5 months, 6 months, 7 months, 8 months, 9 months, 10 months, 11 months, 1 year, 13 months, 14 months, 15 months, 16 months, 17 months, 18 months, 19 months, 20 months, 21 months, 22 months, 23 months, 2 years, 3 years, 4 years, 5 years, 6 years, 7 years, 8 years, 9 years, 10 years, 11 years, 12 years, 13 years, 14 years, 15 years, 16 years, 17 years, 18 years, 19 years, 20 years, 21 years, 22 years, 23 years, 24 years, 25 years, 26 years, 27 years, 28 years, 29 years, 30 years, 31 years, 32 years, 33 years, 34 years, 35 years, 36 years, 37 years, 38 years, 39 years, 40 years, 41 years, 42 years, 43 years, 44 years, 45 years, 46 years, 47 years, 48 years, 49 years, 50 years, 55 years, 60 years, 65 years, or more than 65 years. Expression of the payload may be for 1-5 hours, 1-12 hours, 1-2 days, 1-5 days, 1-2 weeks, 1-3

weeks, 1-4 weeks, 1-2 months, 1-4 months, 1-6 months, 2-6 months, 3-6 months, 3-9 months, 4-8 months, 6-12 months, 1-2 years, 1-5 years, 2-5 years, 3-6 years, 3-8 years, 4-8 years or 5-10 years or 10-15 years, or 15-20 years, or 20-25 years, or 25-30 years, or 30-35 years, or 35-40 years, or 40-45 years, or 45-50 years, or 50-55 years, or 55-60 years, or 60-65 years.

[0071] An AAV vector can comprise a genome of a helper virus. Helper virus proteins are required for the assembly of a recombinant AAV (rAAV), and packaging of a transgene containing a heterologous nucleic acid into the rAAV. The helper virus genes are adenovirus genes E4, E2a and VA, that when expressed in the cell, assist with AAV replication. In some embodiments, an AAV vector comprises E2. In some embodiments, an AAV vector comprises E4. In some embodiments, an AAV vector comprises VA. In some instances, the AAV vector comprises one of helper virus proteins, or any combination.

[0072] The target gene or gene expression product for use in a transgene can be selected from ATP1A2, CACNA1A, SETD5, SHANK3, NF2, DNMT1, TCF4, RAI1, PEX1, ARSA, EIF2B5, EIF2B1, EIF2B2, NPC1, ADAR, MFSD8, STXBPI, PRICKLE2, PRRT2, IDUA, STX1B, Sarcoglycan Alpha (SGCA), glutamic acid decarboxylase 65 (GAD65), glutamic acid decarboxylase 67 (GAD67), CLN2, Nerve Growth Factor (NGF), glial cell derived neurotrophic factor (GDNF), Survival Of Motor Neuron 1, STXBP1, Telomeric (SMNI), Factor X (FIX), Retinoid Isomerohydrolase (RPE65), sarco/endoplasmic reticulum Ca²⁺-ATPase (SERCA2a), Glucocerebrosidase (GCase), galactocerebrosidase (GALC), CDKL5, Frataxin (FXN), Huntingtin (HTT), methyl-CpG binding protein 2 (MECP2), a peroxisomal biogenesis factor (PEX), progranulin (GRN), an antitubulin agent, copper-zinc superoxide dismutase (SOD1), iduronate 2 sulfatase (hIDS), Glucosylceramidase Beta (GBA), fragile X mental retardation 1 (FMR1), NPC Intracellular Cholesterol Transporter 1 (NPC1), SCN1A, C9orf72, NPS3 and a NLRP3 inflammasome. In some embodiments, the peroxisomal biogenesis factor (PEX) is selected from PEX1, PEX2, PEX3, PEX4, PEX5, PEX6, PEX7, PEX10, PEX11B, PEX12, PEX13, PEX14, PEX16, PEX19, and PEX26.

[0073] An AAV vector can comprise a viral genome comprising a nucleic acid encoding the recombinant AAV (rAAV) capsid protein described herein. The viral genome can comprise a Replication (Rep) gene encoding a Rep protein, and Capsid (Cap) gene encoding an AAP protein in the first open reading frame (ORF1) or a Cap protein in the second open reading frame (ORF2). The Rep protein is selected from Rep78, Rep68, Rep52, and Rep40. In some instances, the Cap gene is modified encoding a modified AAV capsid protein described herein. A wild-type Cap gene encodes three proteins, VP1, VP2, and VP3. In some cases, VP1 is modified. In some cases, VP2 is modified. In some cases, VP3 is modified. In some cases, all three VP1-VP3 are modified. The AAV vector can comprise nucleic acids encoding wild-type Rep78, Rep68, Rep52, Rep40 and AAP proteins.

[0074] In some instances, the AAV9 VP1 gene provided in SEQ ID NO: 86 shown in Table 3, is modified to include any one of SEQ ID NOS: 67-85 in FIG. 1. The AAV vector described herein may be used to produce a variant AAV capsid by the methods described herein.

TABLE-US-00005 TABLE 3 VP1 Capsid Protein Nucleic Acid Sequences SEQ ID NO: Identifier Sequence 86 AAV9 > AY530579.1 ATGGCTGCCGATGGTTATCTTCCAGAT Adeno-associated TGGCTCGAGGACAACCTTAGTGAAGG virus 9 isolate hu. 14 AATTCGCGAGTGGTGGGCTTTGAAACC capsid protein VP1 TGGAGCCCCCTCAACCCAAGGCAAATC (cap) gene, complete AACACATCAAGACAACGCTCGAGGT cds CTTGTGCTTCCGGGTACAAATACCTT GGACCCGGCAACGGACTCGACAAGGG GGAGCCGGTCAACGCAGCAGACGCGG CGGCCCTCGAGCACGACAAGGCCTAC GACCAGCAGCTCAAGGCCGGAGACAA CCGGTACCTCAAGTACAACCACGCCGA CGCCGAGTTCCAGGAGCGGCTCAAAG AAGATACGTCTTTTGGGGGCAACCTCG GGCGAGCAGTCTTCCAGGCCAAAAAG AGGCTTCTTGAACCTCTTGGTCTGGTT GAGGAAGCGGCTAAGACGGCTCCTGG

AAAGAGAGCTGTAGACCTGTCT CTAGGGAACCGGACTCCTCCGCGGGTA
TTGGCAAATCGGGTGCACAGCCCGCTA AAAAGAGACTCAATTTTCGGTCAGACTG
GCGACACAGAGTCAGTCCCAGACCCTC AACCAATCGGAGAACCTCCCGCAGCC
CCCTCAGGTGTGGGATCTCTTACAATG GCTTCAGGTGGTGGCGCACCAGTGGCA
GACAATAACGAAGGTGCCGATGGAGT GGGTAGTTCCTCGGGAAATTGGCATTG
CGATTCCCAATGGCTGGGGGACAGAG TCATCACCACCAGCACCCGAACCTGGG
CCCTGCCCACCTACAACAATCACCTCT ACAAGCAAATCTCCAACAGCACATCTG
GAGGATCTTCAAATGACAACGCCTACT TCGGCTACAGCACCCCCTGGGGGTATT
TTGACTTCAACAGATTCCACTGCCACT TCTCACCACGTGACTGGCAGCGACTCA
TCAACAACAACCTGGGGATTCCGGCCTA AGCGACTCAACTTCAAGCTCTTCAACA
TTCAGGTCAAAGAGGTTACGGACAAC AATGGAGTCAAGACCATCGCCAATAA
CCTTACCAGCACGGTCCAGGTCTTCAC GGACTCAGACTATCAGCTCCCGTACGT
GCTCGGGTTCGGCTCACGAGGGCTGCCT CCCGCCGTTCCCAGCGGACGTTTTTCAT
GATTCCTCAGTACGGGTATCTGACGCT TAATGATGGAAGCCAGGCCGTGGGTC
GTTTCGTCTTTTACTGCCTGGAATATTT CCCGTCGCAAATGCTAAGAACGGGTA
ACAACTTCCAGTTCAGCTACGAGTTTG AGAACGTACCTTTCCATAGCAGCTACG
CTCACAGCCAAAGCCTGGACCGACTA ATGAATCCACTCATCGACCAATACTTG
TACTATCTCTCAAAGACTATTAACGGT TCTGGACAGAATCAACAAACGCTAAA
ATTCAGTGTGGCCGGACCCAGCAACAT GGCTGTCCAGGGAAGAACTACATAC
CTGGACCCAGCTACCGACAACAACGT GTCTCAACCACTGTGACTCAAAACAAC
AACAGCGAATTTGCTTGGCCTGGAGCT TCTTCTTGGGCTCTCAATGGACGTAAT
AGCTTGATGAATCCTGGACCTGCTATG GCCAGCCACAAAGAAGGAGAGGACCG
TTTCTTTCTTTTGTCTGGATCTTTAATT TTTGGCAAACAAGGAACTGGAAGAGA
CAACGTGGATGCGGACAAAGTCATGA TAACCAACGAAGAAGAAATTAAAACT
ACTAACCCGGTAGCAACGGAGTCTTAT GGACAAGTGGCCACAAACCACCAGAG
TGCCCAAGCACAGGCGCAGACCGGCT GGGTTCAAAACCAAGGAATACTTCCG
GGTATGGTTTGGCAGGACAGAGATGT GTACCTGCAAGGACCCATTTGGGCCAA
AATTCCTCACACGGACGGCAACTTTCA CCCTTCTCCGCTGATGGGAGGGTTTGG
AATGAAGCACCCGCCTCCTCAGATCCT CATCAAAAACACACCTGTACCTGCGGA
TCCTCCAACGGCCTTCAACAAGGACAA GCTGAACTCTTTCATCACCCAGTATTC
TACTGGCCAAGTCAGCGTGGAGATCG AGTGGGAGCTGCAGAAGGAAAACAGC
AAGCGCTGGAACCCGGAGATCCAGTA CACTTCCAACCTATTACAAGTCTAATAA
TGTTGAATTTGCTGTTAATACTGAAGG TGTATATAGTGAACCCCGCCCCATTGG
CACCAGATACCTGACTCGTAATCTGTAA

Methods of Producing rAAVs

[0075] Disclosed herein are methods of producing the AAV capsids comprising the AAV capsid proteins and viral vector encoding a therapeutic nucleic acid. The AAV capsid proteins are produced by introducing into a cell (e.g., immortalized stem cell) a first vector containing a transgene cassette flanked by inverted terminal repeat (ITR) sequences from a parental AAV virus (the transgene cassette has a promoter sequence that drives transcription of a heterologous nucleic acid in the nucleus of the target cell), a second vector encoding the AAV genome with a AAV capsid protein (encoding the AAV Rep gene as well as the modified Cap gene for the variant being produced), and a third vector encoding helper virus proteins, required for assembly of the AAV capsid structure and packaging of the transgene in the modified AAV capsid structure. The assembled AAV capsid can be isolated and purified from the cell using suitable methods known in the art. Table 1 provide DNA sequences for using in the methods described herein.

[0076] The transgenes contained in a recombinant AAV (rAAV) vector and encapsidated by the AAV capsid proteins of the present disclosure are also provided herein. The transgenes disclosed herein are delivered to a subject for a variety of purposes, such as to treat a disease or condition in the subject. The transgene can be gene editing components that modulate the activity or expression

of a target gene or gene expression product. Alternatively, the transgene is a gene encoding a therapeutic gene expression product that is effective to modulate the activity or expression of itself, or another target gene or gene expression product.

[0077] Aspects disclosed herein provide methods of manufacturing rAAV virus or virus particles comprising: (a) introducing into a cell a nucleic acid comprising: (i) first vector containing a transgene cassette flanked by inverted terminal repeat (ITR) sequences from a parental AAV virus (the transgene cassette has a promoter sequence that drives transcription of a heterologous nucleic acid in the nucleus of the target cell); (ii) a second vector encoding the AAV genome with a AAV capsid protein of the present invention; and (iii) a vector encoding helper virus proteins, required for assembly of the AAV capsid structure and packaging of the transgene in the modified AAV capsid structure; (b) expressing in the cell the AAV capsid protein described herein; (c) assembling an AAV particle comprising the AAV capsid proteins disclosed herein; and (d) packaging the AAV particle. In some instances, the cell is mammalian. In some instances, the cell is immortalized. In some instances, the immortalized cell is an embryonic stem cell. In some instances, the embryonic stem cell is a human embryonic stem cell. In some instances, the human embryonic stem cell is a human embryonic kidney 293 (HEK-293) cell. In some instances, the Cap gene is derived from the deoxyribose nucleic acid (DNA) provided in SEQ ID NO: 86. In some instances, the 5' ITR and the 3' ITR are derived from an AAV2 serotype. In some instances, the 5' ITR and the 3' ITR are derived from an AAV5 serotype. In some instances, the 5' ITR and the 3' ITR are derived from an AAV9 serotype. In some instances, the first nucleic acid sequence and the second nucleic acid sequence are in trans. In some instances, the first nucleic acid sequence and the second nucleic acid sequence are in cis. In some instances, the first nucleic acid sequence, the second nucleic acid sequence and the third nucleic acid sequence, are in trans.

[0078] The Cap gene disclosed here comprises any one of SEQ ID NOS: 67-85 from FIG. 1, which are DNA sequences encoding the modified AAV capsid protein portions of the present disclosure.

[0079] In some instances, the methods comprise packing the first nucleic acid sequence encoding the therapeutic gene expression product such that it becomes encapsidated by the modified AAV capsid protein. In some embodiments, the rAAV particles are isolated, concentrated, and purified using suitable viral purification methods, such as those described herein.

[0080] In some cases, rAAVs of the present disclosure are generated using the methods described in Challis, R. C. et al. Nat. Protoc. 14, 379 (2019). Briefly, triple transfection of HEK293T cells (ATCC) using polyethylenimine (PEI) is performed, viruses are collected after 120 hours from both cell lysates and media and purified over iodixanol. In a non-limiting example, the rAAVs are generated by triple transfection of precursor cells (e.g., HEK293T) cells using a standard transfection protocol (e.g., PEI). Viral particles are harvested from the media after a period of time (e.g., 72 h post transfection) and from the cells and media at a later point in time (e.g., 120 h post transfection). Virus present in the media is concentrated by precipitation with 8% polyethylene glycol (PEG) and 500 mM sodium chloride and the precipitated virus is added to the lysates prepared from the collected cells. The viruses are purified over iodixanol (Optiprep, Sigma) step gradients (15%, 25%, 40% and 60%). Viruses are concentrated and formulated in PBS. Virus titers are determined by measuring the number of DNaseI-resistant vector genome copies (VGs) using qPCR and the linearized genome plasmid as a control.

[0081] The cell can be selected from a human, a primate, a murine, a feline, a canine, a porcine, an ovine, a bovine, an equine, an epine, a caprine and a lupine host cell. In some instances, the cell is a progenitor or precursor cell, such as a stem cell. In some instances, the stem cell is a mesenchymal cell, embryonic stem cell, induced pluripotent stem cell (iPSC), fibroblast or other tissue specific stem cell. The cell can be immortalized. In some cases, the immortalized cell is a HEK293cell. In some instances, the cell is a differentiated cell. Based on the disclosure provided, it is expected that this system can be used in conjunction with any transgenic line expressing a recombinase in the target cell type of interest to develop AAV capsids that more efficiently transduce that target cell

population.

Methods of Treatment

[0082] Disclosed herein are methods of treating a disease or condition, or a symptom of the disease or condition, in a subject, comprising administering of therapeutically effective amount of one or more compositions (e.g., rAAV particle, AAV vector, pharmaceutical composition) disclosed herein to the subject. In some embodiments, the composition is a rAAV capsid protein described herein. In some embodiments, the composition is an isolated and purified rAAV capsid protein described herein. In some embodiments, the rAAV particle encapsidates an AAV vector comprising a transgene (e.g., therapeutic nucleic acid). In some embodiments, the composition is a rAAV capsid protein described herein conjugated with a therapeutic agent disclosed herein. In some embodiments, the composition is a pharmaceutical composition comprising the rAAV particle and a pharmaceutically acceptable carrier. In some embodiments, the one or more compositions are administered to the subject alone (e.g., stand-alone therapy). In some embodiments, the composition is a first-line therapy for the disease or condition. In some embodiments, the composition is a second-line, third-line, or fourth-line therapy, for the disease or condition.

[0083] Recombinant adeno-associated virus (rAAV) mediated gene delivery leverages the AAV mechanism of viral transduction for nuclear expression of an episomal heterologous nucleic acid (e.g., a transgene, therapeutic nucleic acid). Upon delivery to a host in vivo environment, a rAAV will (1) bind or attach to cellular surface receptors on the target cell, (2) endocytose, (3) traffic to the nucleus, (4) uncoat the virus to release the encapsidated heterologous nucleic acid, (5) convert of the heterologous nucleic acid from single-stranded to double-stranded DNA as a template for transcription in the nucleus, and (6) transcribe of the episomal heterologous nucleic acid in the nucleus of the host cell ("transduction"). rAAVs engineered to have an increased transduction enrichment (transcription of the episomal heterologous nucleic acid in the host cell) are desirable for gene therapy applications.

[0084] Aspects disclosed herein provide methods of treating a disease or condition in a subject, the method comprising administering to the subject a therapeutically effective amount of the rAAV of the present disclosure, or the pharmaceutical formulation of the present disclosure, wherein the gene product is a therapeutic gene product. In some embodiments, the administering is by intracranial, intraventricular, intracerebroventricular, intravenous, intraarterial, intranasal, intrathecal, intracisternae magna, or subcutaneous.

[0085] Provided here, are methods of treating a disease or a condition associated with an aberrant expression or activity of a target gene or gene expression product thereof, the method comprising modulating the expression or the activity of a target gene or gene expression product in a subject by administering a rAAV encapsidating a heterologous nucleic acid of the present disclosure. In some instances, the expression or the activity of the target gene or gene expression product is decreased, relative to that in a normal (non-diseased) individual; and administering the rAAV to the subject is sufficient to increase the expression of the activity of the target gene or gene expression product. In some instances, the expression or the activity of the gene or gene expression product is increased, relative to that in a normal individual; and administering the rAAV to the subject is sufficient to decrease the expression or the activity of the target gene or gene expression product. In a non-limiting example, a subject diagnosed with Alzheimer's disease, which is caused, in some cases, by a gain-of-function of a Presenilin 1 and/or Presenilin 2 (encoded by the gene PSEN1 and PSEN2, respectively) is administered a rAAV disclosed herein encapsidating a therapeutic nucleic acid that is a silencing RNA (siRNA), or other RNAi with a loss-of-function effect on PSEN1 mRNA.

[0086] Also provided are methods of preventing a disease or condition disclosed herein in a subject comprising administering to the subject a therapeutically effective amount of an rAAV vector comprising a nucleic acid sequence encoding a therapeutic gene expression product described herein. The rAAV vector may be encapsidated in the modified capsid protein or rAAV viral particle described herein. In some instances, the therapeutic gene expression product is effective to

modulate the activity or expression of a target gene or gene expression product.

[0087] Disclosed herein are methods of treating a disease or condition in a subject by administering a composition comprising a rAAV disclosed herein. An advantage of the rAAVs disclosed herein, is that the rAAV may be used to treat virtually any disease or condition that would benefit from a transgene therapy, including but not limited to spinal muscular atrophy (SMA), amyotrophic lateral sclerosis (ALS), Parkinson's disease, Pompe disease, mucopolysaccharidosis type II, fragile X syndrome, STXBPI encephalopathy, Krabbe disease, Huntington's disease, Alzheimer's disease, Batten's disease, lysosomal storage disorders, glioblastoma multiforme, Rett syndrome, Leber's congenital amaurosis, Late infantile neuronal ceroid lipofuscinosis (LINCL), chronic pain, stroke, spinal cord injury, traumatic brain injury and lysosomal storage disorders.

[0088] In some cases, the disease or condition is localized to a particular in vivo environment in the subject, e.g., the CNS. The compositions of the present disclosure are particularly useful for the treatment of the diseases or conditions described herein because they specifically or more efficiently target the in vivo environment and deliver a therapeutic nucleic acid engineered to modulate the activity or the expression of a target gene expression product involved with the pathogenesis or pathology of the disease or condition.

[0089] Provided herein are methods of treating a disease or a condition, or a symptom of the disease or condition, in a subject, comprising: (a) diagnosing a subject with a disease or a condition affecting a target in vivo environment; and (b) treating the disease or the condition by administering to the subject a therapeutically effective amount of a composition disclosed herein (e.g., rAAV particle, AAV vector, pharmaceutical composition), wherein the composition is engineered with an increased expression.

[0090] Disclosed herein are methods of treating a disease or a condition, or a symptom of the disease or the condition, afflicting a target in a subject comprising: (a) administering to the subject a composition (e.g., rAAV particle, AAV vector, pharmaceutical composition); and (b) expressing the therapeutic nucleic acid into a target in vivo environment in the subject with an increased transduction enrichment.

[0091] In some embodiments, methods further comprise reducing or ablating delivery of the heterologous nucleic acid in an off-target in vivo environment, such as the liver. In some embodiments, delivery is characterized by an increase in enrichment of transduction (e.g., of the heterologous nucleic acid) in the CNS.

[0092] In some embodiments, methods of treating a disease or condition affecting the CNS comprise administering a rAAV particle to a CNS in a subject, the rAAV particle comprising an rAAV capsid protein comprising an insertion of about, five, six, or seven amino acids of an amino acid sequence provided in Table 1 and FIG. 1, at an amino acid position 588-589 in a parental AAV capsid protein. In some embodiments, methods of treating a disease or condition affecting the CNS comprise administering a rAAV particle to a CNS in a subject, the rAAV particle comprising an rAAV capsid protein comprising an insertion of about, five, six, or seven amino acids of an amino acid sequence as well as one or more substitution at amino acid found at amino acid positions 587-590 [AQAQ] such as provided in Table 1 and FIG. 1. In some embodiments, the parental AAV capsid protein is AAV9 capsid protein (for e.g., provided in SEQ ID NO: 1).

[0093] Also provided are methods of modulating a target gene expression product, the methods comprising administering to a subject in need thereof a composition (e.g., rAAV particle, AAV vector, pharmaceutical composition) disclosed herein. For example, methods provided herein comprise administering to a subject a rAAV with a rAAV capsid protein encapsidating a viral vector comprising a heterologous nucleic acid that modulates the expression or the activity of the target gene expression product.

[0094] The term "normal individual" refers to an individual that is not afflicted with the disease or the condition characterized by the variation in expression or activity of the gene or gene expression product thereof.

[0095] In some embodiments, the disease or condition of the CNS is selected from Absence of the Septum Pellucidum, Acid Lipase Disease, Acid Maltase Deficiency, Acquired Epileptiform Aphasia, Acute Disseminated Encephalomyelitis, Attention Deficit-Hyperactivity Disorder (ADHD), Adie's Pupil, Adie's Syndrome, Adrenoleukodystrophy, Agenesis of the Corpus Callosum, Agnosia, Aicardi Syndrome, Aicardi-Goutieres Syndrome Disorder, AIDS-Neurological Complications, Alexander Disease, Alpers' Disease, Alternating Hemiplegia, Alzheimer's Disease, Amyotrophic Lateral Sclerosis (ALS), Anencephaly, Aneurysm, Angelman Syndrome, Angiomatosis, Anoxia, Antiphospholipid Syndrome, Aphasia, Apraxia, Arachnoid Cysts, Arachnoiditis, Arnold-Chiari Malformation, Arteriovenous Malformation, Asperger Syndrome, Ataxia, Ataxia Telangiectasia, Ataxias and Cerebellar or Spinocerebellar Degeneration, Atrial Fibrillation and Stroke, Attention Deficit-Hyperactivity Disorder, Autism Spectrum Disorder, Autonomic Dysfunction, Back Pain, Barth Syndrome, Batten Disease, Becker's Myotonia, Behcet's Disease, Bell's Palsy, Benign Essential Blepharospasm, Benign Focal Amyotrophy, Benign Intracranial Hypertension, Bernhardt-Roth Syndrome, Binswanger's Disease, Blepharospasm, Bloch-Sulzberger Syndrome, Brachial Plexus Birth Injuries, Brachial Plexus Injuries, Bradbury-Eggleston Syndrome, Brain and Spinal Tumors, Brain Aneurysm, Brain Injury, Brown-Sequard Syndrome, Bulbospinal Muscular Atrophy, Cerebral Autosomal Dominant Arteriopathy with Subcortical Infarcts and Leukoencephalopathy (CADASIL), Canavan Disease, Carpal Tunnel Syndrome, Causalgia, Cavernomas, Cavernous Angioma, Cavernous Malformation, Central Cervical Cord Syndrome, Central Cord Syndrome, Central Pain Syndrome, Central Pontine Myelinolysis, Cephalic Disorders, Ceramidase Deficiency, Cerebellar Degeneration, Cerebellar Hypoplasia, Cerebral Aneurysms, Cerebral Arteriosclerosis, Cerebral Atrophy, Cerebral Beriberi, Cerebral Cavernous Malformation, Cerebral Gigantism, Cerebral Hypoxia, Cerebral Palsy, Cerebro-Oculo-Facio-Skeletal Syndrome (COFS), Charcot-Marie-Tooth Disease, Charcot-Marie-Tooth syndrome, classical rhizomelic chondrodysplasia punctata (RCDP), Chiari Malformation, Cholesterol Ester Storage Disease, Chorea, Choreoacanthocytosis, Chronic Inflammatory Demyelinating Polyneuropathy (CIDP), Chronic Orthostatic Intolerance, Chronic Pain, Cockayne Syndrome, Cockayne Syndrome Type II, Coffin Lowry Syndrome, Colpocephaly, Coma, Complex Regional Pain Syndrome, Congenital Facial Diplegia, Congenital Myasthenia, Congenital Myopathy, Congenital Vascular Cavernous Malformations, Corticobasal Degeneration, Cranial Arteritis, Craniosynostosis, Cree encephalitis, Creutzfeldt-Jakob Disease, Cumulative Trauma Disorders, Cushing's Syndrome, Cytomegalic Inclusion Body Disease, Cytomegalovirus Infection, Dancing Eyes-Dancing Feet Syndrome, Dandy-Walker Syndrome, Dawson Disease, Deafness, De Morsier's Syndrome, Dejerine-Klumpke Palsy, Dementia, Dementia-Multi-Infarct, Dementia-Semantic, Dementia-Subcortical, Dementia With Lewy Bodies, Dentate Cerebellar Ataxia, Dentatorubral Atrophy, Dermatomyositis, Developmental Dyspraxia, Devic's Syndrome, Diabetic Neuropathy, Diffuse Sclerosis, Dravet Syndrome, Duchenne muscular dystrophy, Dysautonomia, Dysgraphia, Dyslexia, Dysphagia, Dyspraxia, Dyssynergia Cerebellaris Myoclonica, Dyssynergia Cerebellaris Progressiva, Dystonias, Early Infantile Epileptic Encephalopathy, Empty Sella Syndrome, Encephalitis, Encephalitis Lethargica, Encephaloceles, Encephalopathy, Encephalopathy (familial infantile), Encephalotrigeminal Angiomatosis, Epilepsy, Epileptic Hemiplegia, Erb's Palsy, Erb-Duchenne and Dejerine-Klumpke Palsies, Essential Tremor, Extrapontine Myelinolysis, Fabry Disease, Fahr's Syndrome, Fainting, Familial Dysautonomia, Familial Hemangioma, Familial Idiopathic Basal Ganglia Calcification, Familial Periodic Paralysis, Familial Spastic Paralysis, Farber's Disease, Febrile Seizures, Fibromuscular Dysplasia, Fisher Syndrome, Floppy Infant Syndrome, Foot Drop, Fragile X syndrome, Friedreich's Ataxia, Frontotemporal Dementia (FTD), Gaucher Disease, Generalized Gangliosidoses, Gerstmann's Syndrome, Gerstmann-Straussler-Scheinker Disease, Giant Axonal Neuropathy, Giant Cell Arteritis, Giant Cell Inclusion Disease, glioblastoma, Globoid Cell Leukodystrophy, Glossopharyngeal Neuralgia, Glycogen Storage Disease, Guillain-Barre Syndrome, Hallervorden-

Spatz Disease, Head Injury, Headache, Hemifacial Spasm, Hemiplegia
Alterans, Hereditary Neuropathies, Hereditary Spastic Paraplegia, Heredopathia Atactica
Polyneuritiformis, Herpes Zoster, Herpes Zoster Oticus, Hirayama Syndrome, Holmes-Adie
syndrome, Holoprosencephaly, HTLV-1 Associated Myelopathy, Hughes Syndrome, Huntington's
Disease, Hydranencephaly, Hydrocephalus, Hydrocephalus-Normal Pressure, Hydromyelia,
Hypercortisolism, Hypersomnia, Hypertonia, Hypotonia, Hypoxia, Immune-Mediated
Encephalomyelitis, Inclusion Body Myositis, Incontinentia Pigmenti, Infantile Hypotonia, Infantile
Neuroaxonal Dystrophy, Infantile Phytanic Acid Storage Disease, Infantile Refsum Disease (IRD),
Infantile Spasms, Inflammatory Myopathies, Iniencephaly, Intestinal Lipodystrophy, Intracranial
Cysts, Intracranial Hypertension, Isaacs' Syndrome, Joubert Syndrome, Kearns-Sayre Syndrome,
Kennedy's Disease, Kinsbourne syndrome, Kleine-Levin Syndrome, Klippel-Feil Syndrome,
Klippel-Trenaunay Syndrome (KTS), Kliiver-Bucy Syndrome, Korsakoff s Amnesic Syndrome,
Krabbe Disease, Kugelberg-Welander Disease, Kuru, Lambert-Eaton Myasthenic Syndrome,
Landau-Kleffner Syndrome, Lateral Femoral Cutaneous Nerve Entrapment, Lateral Medullary
Syndrome, Learning Disabilities, Leigh's Disease, Lennox-Gastaut Syndrome, Lesch-Nyhan
Syndrome, Leukodystrophy, Levine-Critchley Syndrome, Lewy Body Dementia, Lipid Storage
Diseases, Lipoid Proteinosis, Lissencephaly, Locked-In Syndrome, Lou Gehrig's Disease, Lupus-
Neurological Sequelae, Lyme Disease-Neurological Complications, Machado-Joseph Disease,
Macrencephaly, Maple syrup urine disease, Megalencephaly, Melkersson-Rosenthal Syndrome,
Meningitis, Meningitis and Encephalitis, Menkes Disease, Menkes syndrome, Meralgia
Paresthetica, Metachromatic Leukodystrophy, Microcephaly, Migraine, Miller Fisher Syndrome,
Mini Stroke, Mitochondrial Myopathy, Moebius Syndrome, Monomelic Amyotrophy, Motor
Neuron Diseases, Moyamoya Disease, Mucopolysaccharidoses, Mucopolysaccharidosis,
Mucopolysaccharidosis II, Multi-Infarct Dementia, Multifocal Motor Neuropathy, Multiple
Sclerosis, Multiple System Atrophy, Multiple System Atrophy with Orthostatic Hypotension,
Muscular Dystrophy, Myasthenia-Congenital, Myasthenia Gravis, Myelinoclastic Diffuse
Sclerosis, Myoclonic Encephalopathy of Infants, Myoclonus, Myopathy, Myopathy-Congenital,
Myopathy-Thyrotoxic, Myotonia, Myotonia Congenita, Myotonic dystrophy, Narcolepsy,
Neuroacanthocytosis, Neurodegeneration with Brain Iron Accumulation, Neurofibromatosis,
Neuroleptic Malignant Syndrome, Neurological Complications of AIDS, Neurological
Complications of Lyme Disease, Neurological Consequences of Cytomegalovirus Infection,
Neurological Manifestations of Pompe Disease, Neurological Sequelae Of Lupus, Neuromyelitis
Optica, Neuromyotonia, Neuronal Ceroid Lipofuscinosis, Neuronal Migration Disorders,
Neuropathy-Hereditary, Neurosarcoidosis, Neurosyphilis, Neurotoxicity, Nevus Cavernosus,
Niemann-Pick Disease, O'Sullivan-McLeod Syndrome, Occipital Neuralgia, Ohtahara Syndrome,
Olivopontocerebellar Atrophy, Opsoclonus Myoclonus, Orthostatic Hypotension, Overuse
Syndrome, Pain-Chronic, Pantothenate Kinase-Associated Neurodegeneration, Paraneoplastic
Syndromes, Paresthesia, Parkinson's Disease, Paroxysmal Choreoathetosis, Paroxysmal
Hemicrania, Parry-Romberg, Pelizaeus-Merzbacher Disease, Pena Shokeir II Syndrome, Perineural
Cysts, Periodic Paralysis, Peripheral Neuropathy, Periventricular Leukomalacia, Persistent
Vegetative State, Pervasive Developmental Disorders, Phenylketonuria, Phytanic Acid Storage
Disease, Pick's Disease, Pinched Nerve, *Piriformis* Syndrome, Pituitary Tumors, Polymyositis,
Pompe Disease, Porencephaly, Post-Polio Syndrome, Postherpetic Neuralgia, Postinfectious
Encephalomyelitis, Postural Hypotension, Postural Orthostatic Tachycardia Syndrome, Postural
Tachycardia Syndrome, Prader-Willi syndrome, Primary Dentatum Atrophy, Primary Lateral
Sclerosis, Primary Progressive Aphasia, Prion Diseases, Progressive Hemifacial Atrophy,
Progressive Locomotor Ataxia, Progressive Multifocal Leukoencephalopathy, Progressive
Sclerosing Poliodystrophy, Progressive Supranuclear Palsy, Prosopagnosia, Pseudo-Torch
syndrome, Pseudotuberculosis syndrome, Pseudotumor Cerebri, Psychogenic Movement,
Ramsay Hunt Syndrome I, Ramsay Hunt Syndrome II, Rasmussen's Encephalitis, Reflex

Sympathetic Dystrophy Syndrome, Refsum Disease, Refsum Disease-Infantile, Repetitive Motion Disorders, Repetitive Stress Injuries, Restless Legs Syndrome, Retrovirus-Associated Myelopathy, Rett Syndrome, Reye's Syndrome, Rheumatic Encephalitis, Riley-Day Syndrome, Sacral Nerve Root Cysts, Saint Vitus Dance, Salivary Gland Disease, Sandhoff Disease, Schilder's Disease, Schizencephaly, Seitelberger Disease, Seizure Disorder, Semantic Dementia, Septo-Optic Dysplasia, Severe Myoclonic Epilepsy of Infancy (SMEI), Shaken Baby Syndrome, Shingles, Shy-Drager Syndrome, Sjogren's Syndrome, Sleep Apnea, Sleeping Sickness, Sotos Syndrome, Spasticity, Spina *Bifida*, Spinal Cord Infarction, Spinal Cord Injury, Spinal Cord Tumors, Spinal Muscular Atrophy, Spinocerebellar ataxia, Spinocerebellar Atrophy, Spinocerebellar Degeneration, Steele-Richardson-Olszewski Syndrome, Stiff-Person Syndrome, Striatonigral Degeneration, Stroke, Sturge-Weber Syndrome, STXBPI encephalopathy, Subacute Sclerosing Panencephalitis, Subcortical Arteriosclerotic Encephalopathy, Short-lasting, Unilateral, Neuralgiform (SUNCT) Headache, Swallowing Disorders, Sydenham Chorea, Syncope, Syphilitic Spinal Sclerosis, Syringohydromyelia, Syringomyelia, Systemic Lupus Erythematosus, Tabes *Dorsalis*, Tangier disease, Tardive Dyskinesia, Tarlov Cysts, Tay-Sachs Disease, Temporal Arteritis, Tethered Spinal Cord Syndrome, Thomsen's Myotonia, Thoracic Outlet Syndrome, Thyrotoxic Myopathy, Tic Douloureux, Todd's Paralysis, Tourette Syndrome, Transient Ischemic Attack, Transmissible Spongiform Encephalopathies, Transverse Myelitis, Traumatic Brain Injury, Tremor, Trigeminal Neuralgia, Tropical Spastic Paraparesis, Troyer Syndrome, Tuberous Sclerosis, Vascular Erectile Tumor, Vasculitis Syndromes of the Central Nervous Systems, Von Economo's Disease, Von Hippel-Lindau Disease (VHL), Von Hippel-Lindau syndrome, Von Recklinghausen's Disease, Wallenberg's Syndrome, Werdnig-Hoffman Disease, Wernicke-Korsakoff Syndrome, West Syndrome, Whiplash, Whipple's Disease, Williams Syndrome, Wilson Disease, Wolman's Disease, X-Linked Spinal and Bulbar Muscular Atrophy and Zellweger syndrome.

[0096] In some embodiments, the pharmaceutical formulation comprises a therapeutic nucleic acid encoding a therapeutic gene expression product. In some instances, the therapeutic gene expression product is effective to modulate an activity or an expression of a target gene or gene expression product selected from ATP1A2, CACNA1A, SETD5, SHANK3, NF2, DNMT1, TCF4, RAI1, PEX1, ARSA, EIF2B5, EIF2B1, EIF2B2, NPC1, ADAR, MFSD8, STXBPI, PRICKLE2, PRRT2, IDUA, STX1B, Sarcoglycan Alpha (SGCA), glutamic acid decarboxylase 65 (GAD65), glutamic acid decarboxylase 67 (GAD67), CLN2, Nerve Growth Factor (NGF), glial cell derived neurotrophic factor (GDNF), Survival Of Motor Neuron 1, STXBP1, Telomeric (SMNI), Factor X (FIX), Retinoid Isomerohydrolase (RPE65), sarco/endoplasmic reticulum Ca²⁺-ATPase (SERCA2a), Glucocerebrosidase (GCase), galactocerebrosidase (GALC), CDKL5, Frataxin (FXN), Huntingtin (HTT), methyl-CpG binding protein 2 (MECP2), a peroxisomal biogenesis factor (PEX), progranulin (GRN), an antitubulin agent, copper-zinc superoxide dismutase (SOD1), iduronate 2 sulfatase (hIDS), Glucosylceramidase Beta (GBA), fragile X mental retardation 1 (FMR1), NPC Intracellular Cholesterol Transporter 1 (NPC1), SCN1A, C9orf72, NPS3 and a NLRP3 inflammasome. In some embodiments, the peroxisomal biogenesis factor (PEX) is selected from PEX1, PEX2, PEX3, PEX4, PEX5, PEX6, PEX7, PEX10, PEX11B, PEX12, PEX13, PEX14, PEX16, PEX19, and PEX26.

[0097] In some aspects, other examples of genes involved in CNS diseases or disorders include MAPT, IDUA, SNCA, ATXN2, Ube3a, GNS, HGSNAT, NAGLU, SGSH, CLN1, CLN3, CLN4, CLN5, CLN6, CLN7, CLN8, CTSD, ABCD1, HEXA, HEXB, ASM, ASPA, GLB1, AADC, MFN2, GNAO1, SYNGAP1, GRIN2A, GRIN2B, KCNQ2, EPM2A, NHLRC1, SLC6A1, SLC13A5, SURF1, GBE1, ATXN1, ATXN3, and ATXN7.

[0098] In some instances, the therapeutic gene expression product comprises gene editing components. In some instances, the gene editing components are selected from an artificial site-specific RNA endonuclease (ASRE), a zinc finger endonuclease (ZFN), a transcription factor like effector nuclease (TALEN), a clustered regularly interspaced short palindromic repeats

(CRISPR)/Cas enzyme, and a CRISPR)/Cas guide RNA.

[0099] In some instances, the expression of a gene or expression or activity of a gene expression product is inhibited by the administration of the composition to the subject. In some instances, the expression of a gene or the expression or the activity of a gene expression product is enhanced by the administration of the composition to the subject.

Formulations, Dosages, and Routes of Administration

[0100] Disclosed herein are methods comprising delivering a rAAV particle encapsidating a heterologous nucleic acid to the CNS in a subject, the rAAV particle comprising (i) an increased transduction of the heterologous nucleic acid in the CNS, wherein the rAAV particle has an rAAV capsid protein comprising an insertion of five, six, or seven amino acids of an amino acid sequence provided in Table 1 and FIG. 1, at an amino acid position 588-589 in a parental AAV capsid protein or comprising an rAAV capsid protein comprising an insertion of about, five, six, or seven amino acids of an amino acid sequence as well as one or more substitution at amino acid found at amino acid positions 587-590 [AQAQ] such as provided in Table 1 and FIG. 1.

[0101] In general, methods disclosed herein comprise administering a therapeutic rAAV composition by systemic administration. In some instances, methods comprise administering a therapeutic rAAV composition by intravenous (“i.v.”) administration. One may administer therapeutic rAAV compositions by additional routes, such as subcutaneous injection, intramuscular injection, intradermal injection, transdermal injection, percutaneous administration, intranasal administration, intralymphatic injection, rectal administration intragastric administration, intraocular administration, intracerebroventricular administration, intrathecally, intracisternal, or any other suitable parenteral administration. Routes, dosage, time points, and duration of administering therapeutics may be adjusted. In some embodiments, administration of therapeutics is prior to, or after, onset of either, or both, acute and chronic symptoms of the disease or condition. Other routes of delivery to the CNS include, but are not limited to intracranial administration, lateral cerebroventricular administration, and endovascular administration.

[0102] An effective dose and dosage of pharmaceutical compositions to prevent or treat the disease or condition disclosed herein is defined by an observed beneficial response related to the disease or condition, or symptom of the disease or condition. Beneficial response comprises preventing, alleviating, arresting, or curing the disease or condition, or symptom of the disease or condition. In some embodiments, the beneficial response may be measured by detecting a measurable improvement in the presence, level, or activity, of biomarkers, transcriptomic risk profile, or intestinal microbiome in the subject. An “improvement,” as used herein refers to shift in the presence, level, or activity towards a presence, level, or activity, observed in normal individuals (e.g., individuals who do not suffer from the disease or condition). In instances wherein the therapeutic rAAV composition is not therapeutically effective or is not providing a sufficient alleviation of the disease or condition, or symptom of the disease or condition, then the dosage amount and/or route of administration may be changed, or an additional agent may be administered to the subject, along with the therapeutic rAAV composition. In some embodiments, as a patient is started on a regimen of a therapeutic rAAV composition, the patient is also weaned off (e.g., step-wise decrease in dose) a second treatment regimen.

[0103] In some cases, a dose of the pharmaceutical composition may comprise a concentration of infectious particles of at least or about $10^{sup.7}$, $10^{sup.8}$, $10^{sup.9}$, $10^{sup.10}$, $10^{sup.11}$, $10^{sup.12}$, $10^{sup.13}$, $10^{sup.14}$, $10^{sup.15}$, $10^{sup.16}$, or $10^{sup.17}$. In some cases, the concentration of infectious particles is $2 \times 10^{sup.7}$, $2 \times 10^{sup.8}$, $2 \times 10^{sup.9}$, $2 \times 10^{sup.10}$, $2 \times 10^{sup.11}$, $2 \times 10^{sup.12}$, $2 \times 10^{sup.13}$, $2 \times 10^{sup.14}$, $2 \times 10^{sup.15}$, $2 \times 10^{sup.16}$, or $2 \times 10^{sup.17}$. In some cases, the concentration of the infectious particles is $3 \times 10^{sup.7}$, $3 \times 10^{sup.8}$, $3 \times 10^{sup.9}$, $3 \times 10^{sup.10}$, $3 \times 10^{sup.11}$, $3 \times 10^{sup.12}$, $3 \times 10^{sup.13}$, $3 \times 10^{sup.14}$, $3 \times 10^{sup.15}$, $3 \times 10^{sup.16}$, or $3 \times 10^{sup.17}$. In some cases, the concentration of the infectious particles is $4 \times 10^{sup.7}$, $4 \times 10^{sup.8}$, $4 \times 10^{sup.9}$, $4 \times 10^{sup.10}$, $4 \times 10^{sup.11}$, $4 \times 10^{sup.12}$, $4 \times 10^{sup.13}$, $4 \times 10^{sup.14}$, $4 \times 10^{sup.15}$,

4×10^{sup.16}, or 4×10^{sup.17}. In some cases, the concentration of the infectious particles is 5×10^{sup.7}, 5×10^{sup.8}, 5×10^{sup.9}, 5×10^{sup.10}, 5×10^{sup.11}, 5×10^{sup.12}, 5×10^{sup.13}, 5×10^{sup.14}, 5×10^{sup.15}, 5×10^{sup.16}, or 5×10^{sup.17}. In some cases, the concentration of the infectious particles is 6×10^{sup.7}, 6×10^{sup.8}, 6×10^{sup.9}, 6×10^{sup.10}, 6×10^{sup.11}, 6×10^{sup.12}, 6×10^{sup.13}, 6×10^{sup.14}, 6×10^{sup.15}, 6×10^{sup.16}, or 6×10^{sup.17}. In some cases, the concentration of the infectious particles is 7×10^{sup.7}, 7×10^{sup.8}, 7×10^{sup.9}, 7×10^{sup.10}, 7×10^{sup.11}, 7×10^{sup.12}, 7×10^{sup.13}, 7×10^{sup.14}, 7×10^{sup.15}, 7×10^{sup.16}, or 7×10^{sup.17}. In some cases, the concentration of the infectious particles is 8×10^{sup.7}, 8×10^{sup.8}, 8×10^{sup.9}, 8×10^{sup.10}, 8×10^{sup.11}, 8×10^{sup.12}, 8×10^{sup.13}, 8×10^{sup.14}, 8×10^{sup.15}, 8×10^{sup.16}, or 8×10^{sup.17}. In some cases, the concentration of the infectious particles is 9×10^{sup.7}, 9×10^{sup.8}, 9×10^{sup.9}, 9×10^{sup.10}, 9×10^{sup.11}, 9×10^{sup.12}, 9×10^{sup.13}, 9×10^{sup.14}, 9×10^{sup.15}, 9×10^{sup.16}, or 9×10^{sup.17}.

[0104] Disclosed herein, in some embodiments are formulations of pharmaceutically-acceptable excipients and carrier solutions suitable for delivery of the rAAV compositions described herein, as well as suitable dosing and treatment regimens for using the particular compositions described herein in a variety of treatment regimens. In some embodiments, the amount of therapeutic gene expression product in each therapeutically-useful composition may be prepared in such a way that a suitable dosage will be obtained in any given unit dose of the compound. Factors such as solubility, bioavailability, biological half-life, route of administration, product shelf life, as well as other pharmacological considerations will be contemplated by one skilled in the art of preparing such pharmaceutical formulations, and as such, a variety of dosages and treatment regimens may be desirable.

[0105] In some embodiments, the pharmaceutical forms of the rAAV-based viral compositions suitable for injectable use include sterile aqueous solutions or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (e.g., glycerol, propylene glycol, and liquid polyethylene glycol, and the like), suitable mixtures thereof, and/or vegetable oils. Proper fluidity may be maintained, for example, by the use of a coating, such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. The prevention of the action of microorganisms can be brought about by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, sorbic acid, thimerosal, and the like. In many cases, it will be preferable to include isotonic agents, for example, sugars or sodium chloride.

[0106] In some cases, for administration of an injectable aqueous solution, the solution may be suitably buffered, if necessary, and the liquid diluent first rendered isotonic with sufficient saline or glucose. These particular aqueous solutions are especially suitable for intravenous, intramuscular, subcutaneous and intraperitoneal administration. Some variation in dosage will necessarily occur depending on the condition of the subject being treated. The person responsible for administration will, in any event, determine the appropriate dose for the individual subject. Moreover, for human administration, preparations should meet sterility, pyrogenicity, and the general safety and purity standards as required by FDA Office of Biologics standards.

[0107] Disclosed herein are sterile injectable solutions comprising the rAAV compositions disclosed herein, which are prepared by incorporating the rAAV compositions disclosed herein in the required amount in the appropriate solvent with several of the other ingredients enumerated above, as required, followed by filtered sterilization. Generally, dispersions are prepared by incorporating the various sterilized active ingredients into a sterile vehicle which contains the basic dispersion medium and the required other ingredients from those enumerated above. In the case of sterile powders for the preparation of sterile injectable solutions, the preferred methods of preparation are vacuum-drying and freeze-drying techniques which yield a powder of the active ingredient plus any additional desired ingredient from a previously sterile-filtered solution thereof.

Injectable solutions may be advantageous for systemic administration, for example by intravenous or intrathecal administration.

[0108] Suitable dose and dosage administered to a subject is determined by factors including, but not limited to, the particular therapeutic rAAV composition, disease condition and its severity, the identity (e.g., weight, sex, age) of the subject in need of treatment, and can be determined according to the particular circumstances surrounding the case, including, e.g., the specific agent being administered, the route of administration, the condition being treated, and the subject or host being treated.

[0109] The amount of rAAV compositions and time of administration of such compositions will be within the purview of the skilled artisan having benefit of the present teachings. It is likely, however, that the administration of therapeutically-effective amounts of the disclosed compositions may be achieved by a single administration, for example, a single injection of sufficient numbers of infectious particles to provide therapeutic benefit to the patient undergoing such treatment. This is made possible, at least in part, by the fact that certain target cells (e.g., neurons) do not divide, obviating the need for multiple or chronic dosing.

[0110] In certain embodiments, the data obtained from cell culture assays and animal studies are used in formulating the therapeutically effective daily dosage range and/or the therapeutically effective unit dosage amount for use in mammals, including humans. In certain embodiments, the dosage range and/or the unit dosage amount varies within this range depending upon the dosage form employed and the route of administration utilized.

Combination Therapies

[0111] A therapeutic rAAV may be used alone or in combination with an additional therapeutic agent (together, “therapeutic agents”). In some cases, a therapeutic rAAV as used herein is administered alone. The therapeutic agent may be administered together or sequentially in a combination therapy. The combination therapy may be administered within the same day, or may be administered one or more days, weeks, months, or years apart.

[0112] The additional therapeutic agent can comprise a small molecule. The additional therapeutic agent can comprise an antibody, or antigen-binding fragment. The additional therapeutic agent can include lipid nanoparticle-based therapies, anti-sense oligonucleotide therapies, as well as other viral therapies.

[0113] The additional therapeutic agent can comprise a cell-based therapy. Exemplary cell-based therapies include without limitation immune effector cell therapy, chimeric antigen receptor T-cell (CAR-T) therapy, natural killer cell therapy and chimeric antigen receptor natural killer (NK) cell therapy. Either NK cells, or CAR-NK cells, or a combination of both NK cells and CAR-NK cells can be used in combination with the methods disclosed herein. In some embodiments, the NK cells and CAR-NK cells are derived from human induced pluripotent stem cells (iPSC), umbilical cord blood, or a cell line. The NK cells and CAR-NK cells can comprise a cytokine receptor and a suicide gene. The cell-based therapy can comprise a stem cell therapy. The stem cell therapy may be embryonic or somatic stem cells. The stem cells may be isolated from a donor (allogeneic) or isolated from the subject (autologous). The stem cells may be expanded adipose-derived stem cells (eASCs), hematopoietic stem cells (HSCs), mesenchymal stem (stromal) cells (MSCs), or induced pluripotent stem cells (iPSCs) derived from the cells of the subject.

Kits

[0114] Disclosed herein are kits comprising compositions disclosed herein. Also disclosed herein are kits for the treatment or prevention of a disease or conditions of the CNS. In some instances, the disease or condition is cancer, a pathogen infection, pulmonary disease or condition, neurological disease, muscular disease, or an immune disorder, such as those described herein.

[0115] In one embodiment, a kit can include a therapeutic or prophylactic composition containing an effective amount of a composition of a rAAV particle encapsidating a recombinant AAV vector encoding a therapeutic nucleic acid (e.g., therapeutic nucleic acid) and a recombinant AAV (rAAV)

capsid protein of the present disclosure. In another embodiment, a kit can include a therapeutic or prophylactic composition containing an effective amount of cells modified by the rAAV described herein (“modified cell”), in unit dosage form that express therapeutic nucleic acid. In some embodiments, a kit comprises a sterile container which can contain a therapeutic composition; such containers can be boxes, ampules, bottles, vials, tubes, bags, pouches, blister-packs, or other suitable container forms known in the art. Such containers can be made of plastic, glass, laminated paper, metal foil, or other materials suitable for holding medicaments.

[0116] In some instances, the kit further comprises a cell. In some instances, the cell is mammalian. In some instances, the cell is immortalized. In some instances, the immortalized cell is an embryonic stem cell. In some instances, the embryonic stem cell is a human embryonic stem cell. In some instances, the human embryonic stem cell is a human embryonic kidney 293 (HEK-293) cell. In some instances, the kit further comprises an AAV vector comprising a heterologous nucleic acid encoding a therapeutic gene expression product. In some instances, the AAV vector is an episome.

[0117] In some cases, rAAV are provided together with instructions for administering the rAAV to a subject having or at risk of developing the disease or condition (e.g., disease of the CNS). Instructions can generally include information about the use of the composition for the treatment or prevention of the disease or condition.

[0118] In some cases, the instructions include at least one of the following: description of the therapeutic rAAV composition; dosage schedule and administration for treatment or prevention of the disease or condition disclosed herein; precautions; warnings; indications; counter-indications; overdosage information; adverse reactions; animal pharmacology; clinical studies; and/or references. The instructions can be printed directly on the container (when present), or as a label applied to the container, or as a separate sheet, pamphlet, card, or folder supplied in or with the container. In some cases, instructions provide procedures for administering the rAAV to the subject alone. In some instances, the instructions provide that the rAAV is formulated for systemic delivery.

Definitions

[0119] The terminology used herein is for the purpose of describing particular cases only and is not intended to be limiting. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Furthermore, to the extent that the terms “including”, “includes”, “having”, “has”, “with”, or variants thereof are used in either the detailed description and/or the claims, such terms are intended to be inclusive in a manner similar to the term “comprising.”

[0120] The term “about” or “approximately” means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, e.g., the limitations of the measurement system. For example, “about” can mean within 1 or more than 1 standard deviation, per the practice in the given value. Where particular values are described in the application and claims, unless otherwise stated the term “about” should be assumed to mean an acceptable error range for the particular value.

[0121] As used herein “consisting essentially of” when used to define compositions and methods, shall mean excluding other elements of any essential significance to the combination for the stated purpose. Thus, a composition consisting essentially of the elements as defined herein would not exclude other materials or steps that do not materially affect the basic and novel characteristic(s) of the claimed disclosure, such as compositions for treating skin disorders like acne, eczema, psoriasis, and rosacea.

[0122] The terms “homologous,” “homology,” or “percent homology” are used herein to generally mean an amino acid sequence or a nucleic acid sequence having the same, or similar sequence to a reference sequence. Percent homology of sequences can be determined using the most recent version of BLAST, as of the filing date of this application.

[0123] The terms “increased,” or “increase” are used herein to generally mean an increase by a statically significant amount. In some embodiments, the terms “increased,” or “increase,” mean an increase of at least 10% as compared to a reference level, for example an increase of at least about 10%, at least about 20%, or at least about 30%, or at least about 40%, or at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90% or up to and including a 100% increase or any increase between 10-100% as compared to a reference level, standard, or control. Other examples of “increase” include an increase of at least 2-fold, at least 5-fold, at least 10-fold, at least 20-fold, at least 50-fold, at least 100-fold, at least 1000-fold or more as compared to a reference level.

[0124] The terms, “decreased” or “decrease” are used herein generally to mean a decrease by a statistically significant amount. In some embodiments, “decreased” or “decrease” means a reduction by at least 10% as compared to a reference level, for example a decrease by at least about 20%, or at least about 30%, or at least about 40%, or at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90% or up to and including a 100% decrease (e.g., absent level or non-detectable level as compared to a reference level), or any decrease between 10-100% as compared to a reference level. In the context of a marker or symptom, by these terms is meant a statistically significant decrease in such level. The decrease can be, for example, at least 10%, at least 20%, at least 30%, at least 40% or more, and is preferably down to a level accepted as within the range of normal for an individual without a given disease.

[0125] The terms “subject” is any organism. In some instances, the organism is a mammal. Non-limiting examples of mammal include, any member of the mammalian class: humans, non-human primates such as chimpanzees, and other apes and monkey species; farm animals such as cattle, horses, sheep, goats, swine; domestic animals such as rabbits, dogs, and cats; laboratory animals including rodents, such as rats, mice and guinea pigs, and the like. In one aspect, the mammal is a human. The term “animal” as used herein comprises human beings and non-human animals. In one embodiment, a “non-human animal” is a mammal, for example a rodent such as rat or a mouse. In one embodiment, a “non-human primate” is a mammal, for example a monkey. In some instances, the subject is a patient, which as used herein, may refer to a subject diagnosed with a particular disease or disorder.

[0126] The term “gene,” as used herein, refers to a segment of nucleic acid that encodes an individual protein or RNA (also referred to as a “coding sequence” or “coding region”), optionally together with associated regulatory region such as promoter, operator, terminator and the like, which may be located upstream or downstream of the coding sequence.

[0127] The term “adeno-associated virus,” or “AAV” as used herein refers to the adeno-associated virus or derivatives thereof. Non-limited examples of AAV's include AAV type 1 (AAV1), AAV type 2 (AAV2), AAV type 3 (AAV3), AAV type 4 (AAV4), AAV type 5 (AAV5), AAV type 6 (AAV6), AAV type 7 (AAV7), AAV type 8 (AAV8), AAV type 9 (AAV9), AAV type 10 (AAV10), AAV type 11 (AAV11), AAV type 12 (AAV12), avian AAV, bovine AAV, canine AAV, equine AAV, primate AAV, non-primate AAV, and ovine AAV. In some instances, the AAV is described as a “Primate AAV,” which refers to AAV that infect primates. Likewise an AAV may infect bovine animals (e.g., “bovine AAV”, and the like). In some instances, the AAV is wildtype, or naturally occurring. In some instances, the AAV is recombinant.

[0128] The term “AAV capsid” as used herein refers to a capsid protein or peptide of an adeno-associated virus. In some instances, the AAV capsid protein is configured to encapsidate genetic information (e.g., a transgene, therapeutic nucleic acid, viral genome). In some instances, the AAV capsid of the instant disclosure is a modified AAV capsid, relative to a corresponding parental AAV capsid protein.

[0129] The term “tropism” as used herein refers to a quality or characteristic of the AAV capsid that may include an increase or a decrease in enrichment of, expressing the encapsidated genetic

information into an in vivo environment, relative to a second in vivo environment. An in vivo environment, in some instances, is a cell-type. An in vivo environment, in some instances, is an organ or organ system.

[0130] The term “AAV vector” as used herein refers to nucleic acid polymer encoding genetic information related to the virus. The AAV vector may be a recombinant AAV vector (rAAV), which refers to an AAV vector generated using recombinatorial genetics methods. In some instances, the rAAV vector comprises at least one heterologous polynucleotide (e.g., a polynucleotide other than a wild-type or naturally occurring AAV genome such as a transgene).

[0131] The term “AAV particle” as used herein refers to an AAV virus, virion, AAV capsid protein or component thereof. In some cases, the AAV particle is modified relative to a parental AAV particle.

[0132] The term “gene product” or “gene expression product” refers to an expression product of a polynucleotide sequence such as, for e.g., a polypeptide, peptide, protein or RNA, including interfering RNA (e.g., siRNA, miRNA, shRNA) and messenger RNA (mRNA).

[0133] The term “heterologous” as used herein refers to a genetic element (e.g., coding region) or gene expression product (e.g., RNA, protein) that is derived from a genotypically distinct entity from that of the rest of the entity to which it is being compared.

[0134] The term “endogenous” as used herein refers to a genetic element (e.g., coding region) or gene expression product (e.g., RNA, protein) that is naturally occurring in or associated with an organism or a particular cell within the organism.

[0135] The terms “treat,” “treating,” and “treatment” as used herein refers to alleviating or abrogating a disorder, disease, or condition; or one or more of the symptoms associated with the disorder, disease, or condition; or alleviating or eradicating a cause of the disorder, disease, or condition itself. Desirable effects of treatment can include, but are not limited to, preventing occurrence or recurrence of disease, alleviation of symptoms, diminishing any direct or indirect pathological consequences of the disease, preventing metastasis, decreasing the rate of disease progression, amelioration or palliation of the disease state and remission or improved prognosis.

[0136] The term “therapeutically effective amount” refers to the amount of a compound or therapy that, when administered, is sufficient to prevent development of, or alleviate to some extent, one or more of the symptoms of a disorder, disease, or condition of the disease; or the amount of a compound that is sufficient to elicit biological or medical response of a cell, tissue, system, animal, or human that is being sought by a researcher, veterinarian, medical doctor, or clinician.

[0137] The term “pharmaceutically acceptable carrier,” “pharmaceutically acceptable excipient,” “physiologically acceptable carrier,” or “physiologically acceptable excipient” refers to a pharmaceutically acceptable material, composition, or vehicle, such as a liquid or solid filler, diluent, excipient, solvent, or encapsulating material. A component can be “pharmaceutically acceptable” in the sense of being compatible with the other ingredients of a pharmaceutical formulation. It can also be suitable for use in contact with the tissue or organ of humans and animals without excessive toxicity, irritation, allergic response, immunogenicity, or other problems or complications, commensurate with a reasonable benefit/risk ratio. See, Remington: The Science and Practice of Pharmacy, 21st Edition; Lippincott Williams & Wilkins: Philadelphia, PA, 2005; Handbook of Pharmaceutical Excipients, 5th Edition; Rowe et al., Eds., The Pharmaceutical Press and the American Pharmaceutical Association: 2005; and Handbook of Pharmaceutical Additives, 3rd Edition; Ash and Ash Eds., Gower Publishing Company: 2007; Pharmaceutical Preformulation and Formulation, Gibson Ed., CRC Press LLC: Boca Raton, FL, 2004).

[0138] The term “pharmaceutical composition” refers to a mixture of a compound disclosed herein with other chemical components, such as diluents or carriers. The pharmaceutical composition can facilitate administration of the compound to an organism. Multiple techniques of administering a compound exist in the art including, but not limited to, systemic administration.

[0139] Non-limiting examples of “sample” include any material from which nucleic acids and/or

proteins can be obtained. As non-limiting examples, this includes whole blood, peripheral blood, plasma, serum, saliva, mucus, urine, semen, lymph, fecal extract, cheek swab, cells or other bodily fluid or tissue, including but not limited to tissue obtained through surgical biopsy or surgical resection. Alternatively, a sample can be obtained through primary patient derived cell lines, or archived patient samples in the form of preserved samples, or fresh frozen samples.

[0140] The term “in vivo” is used to describe an event that takes place in a subject's body.

[0141] The term “in vitro” is used to describe an event that takes places contained in a container for holding laboratory reagent such that it is separated from the biological source from which the material is obtained. In vitro assays can encompass cell-based assays in which living or dead cells are employed. In vitro assays can also encompass a cell-free assay in which no intact cells are employed.

[0142] The term “CNS” or “central nervous system” means a tissue selected from brain, thalamus, cortex, putamen, lateral ventricles, medulla, the pons, the amygdala, the motor cortex, caudate, hypothalamus, striatum, ventral midbrain, neocortex, basal ganglia, hippocampus, cerebrum, cerebellum, brain stem, and spinal cord. The brain includes a variety of cortical and subcortical areas, including the frontal, temporal, occipital and parietal lobes.

[0143] The term “systemic delivery” is defined as a route of administration of medication or other substance into a circulatory system so that the entire body is affected. Administration can take place via enteral administration (absorption of the drug through the gastrointestinal tract) or parenteral administration (generally injection, infusion, or implantation). “Circulatory system” includes both blood or cerebrospinal fluid circulatory systems. Examples of systemic administration for the CNS include intraarterial, intravenous or intrathecal injection. Other examples include administration to the cerebrospinal fluid at any location, in the spine (i.e., but not limited to lumbar) or brain (i.e., but not limited to cisterna magna). The terms “systemic administration” and “systemic delivery” are used interchangeably.

[0144] The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described.

EXAMPLES

Example 1

Method of Identifying the Modified Capsid Proteins in African Green Monkeys

[0145] Of primary concern for the therapeutic applicability of engineered adeno-associated viruses (AAVs) is how well their transduction profiles translate to human application. While previous engineering efforts have focused on in vitro or in vivo rodent screening platforms due to the ease and flexibility of their use, screening efforts directly in non-human primates (NHPs) are much more likely to identify viruses that translate. African Green Monkeys, an old world NHP were selected for the engineering efforts that supported development of the present invention.

Engineering efforts were primarily directed on a region of the AAV9 capsid surface located at amino acid position 588, one of the most exposed loops on the capsid surface that is a variable region between natural AAV serotypes and has a role in receptor binding. Insertion of peptides between positions 588 and 589 has been studied in the past by the present inventors, and others, and has resulted in novel receptor binding (AAV-PHP.B/AAV-PHP.eB binding of Ly6a on rodent brain endothelium to facilitate blood-brain barrier crossing and high transduction of the brain) and drastically altered capsid tropism. A library of viral capsids was created by performing a random 7 amino acid insertion at this site within AAV9 in hopes of achieving a novel tropism toward the NHP CNS.

[0146] Plasmids. The first-round viral DNA library was generated by amplification of a section of the AAV9 capsid genome between amino acids 450-599 using NNK degenerate primers (Integrated DNA Technologies, Inc., IDT) to insert seven random amino acids between amino acids 588 and 589 with all possible variations. The resulting library inserts were then introduced into the rAAV-ACap-in-rev-RNA plasmid via Gibson assembly as previously described. The resulting capsid

DNA library, rAAV-Cap-Cag-GFP11, contained a diversity of ~1.28 billion variants at the amino acid level. The second round viral DNA library was generated similarly to the first round, but instead of NNK degenerate primers inserted at the 588, a synthesized oligo pool (Twist Bioscience) was used to generate only selected variants in a UBC-Cap-DNA and CAG-Cap-DNA construct with CAP. This second-round DNA library contained a diversity from 3000-15,000 variants at the amino acid level with 2-6 barcoded replicates for each variant.

[0147] The AAV2/9 REP-AAP-ACAP plasmid transfected into HEK293T cells to provide the Rep gene for library viral production prevents production of a wild-type AAV9 capsid during viral library production after a plausible recombination event between this plasmid co-transfected with the library plasmids at each stage containing the library inserts.

[0148] Viral production. Recombinant AAVs were generated according to established protocols. Briefly, immortalized HEK293T cells (ATCC) were quadruple transfected with four vectors using polyethylenimine (PEI). The first vector was the rAAV-Cap-in-cis-Lox library flanked by inverted terminal repeat (ITR) sequences from a parental AAV virus. The second vector was the AAV2/9 REP-AAP-ACAP plasmid. The third vector contains nucleic acids encoding helper virus proteins needed for viral assembly and packaging of the heterologous nucleic acid into the modified capsid structure. The fourth is a pUC-18 plasmid included to achieve the right PEI/DNA ratio for optimal transfection enrichment. Only 10 ng of rAAV-Cap-in-cis-Lox library DNA was transfected (per 150 mm plate) to decrease the likelihood of multiple library DNAs entering the same cell. Viral particles are harvested from the cells and media after 60 h post transfection. Virus present in the media is concentrated by precipitation with 8% polyethylene glycol and 500 mM sodium chloride and the precipitated virus is added to the lysates prepared from the collected cells. The viruses are purified over iodixanol (Optiprep, Sigma) step gradients (15%, 25%, 40%, and 60%). Viruses are concentrated and formulated in PBS. Virus titers are determined by measuring the number of DNaseI-resistant vector genome copies (VGs) using qPCR and the linearized genome plasmid as a control.

[0149] Animals. African Green Monkey procedures were approved by the IACUC committee of Virscio, Inc. African Green monkeys were born and raised in Virscio, Inc. colonies and housed under standard conditions. They were fed ad libitum and received enrichment as part of the primate enrichment program for NHPs at Virscio, Inc. For AAV infusions, animals were screened for endogenous neutralizing antibodies (nAb). None of the animals that were screened showed any detectable blocking reaction at 1:5 dilution of serum. The day before infusion the animals' food was removed. The test article was administered via intravenous infusion or lumbar intrathecal dosing. Activity and behavior were monitored daily over the course of in-life.

[0150] DNA/RNA recovery and sequencing. Round 1 and round 2 viral libraries were injected into marmosets at a dose of African Green Monkeys at a dose of between $\kappa \times 10^{\text{sup.12}}$ to $1 \times 10^{\text{sup.13}}$ vg/kg animal and rAAV genomes were recovered three weeks post injection. Animals were euthanized and brain (both round 1 and round 2), spinal cord (round 1 and round 2) and liver (round 1 and round 2) were recovered, snap frozen, and placed into long-term storage at -80°C . as well as other peripheral tissues such as heart, spleen, adrenal, kidney, and quad. For round 1, the brain was separated into eleven brain regions, 20 mg each, and for round 2, eleven brain regions. 20-300 mg of each brain section, spinal cord, and liver was homogenized in buffer using the MagMAX DNA ULTRA (A25597) and a Bead Ruptor 96 (OMNI, INC) and viral DNA was isolated according to the manufacturers recommended protocol. Recovered viral DNA was treated with RNase, and purified with a Zymo DNA Clean and Concentrator kit (D4033). Viral genomes were enriched by 25 cycles of PCR amplification with primers flanking the 588-589 insertion site in the capsid genome using 50% of the total extracted viral DNA as a template. After Zymo DNA purification, samples were diluted 1:10 to 1:1000 depending on tissue type and each dilution further amplified around the library variable region with 10 cycles of PCR. Subsequently, samples were further amplified using custom primers with Illumina Indices for 10 more cycles. The

amplification products were run on a 2% low-melting point agarose gel (ThermoFisher Scientific, 16520050) for better separation and recovery of the 210 bp band.

[0151] For the second round library only, packaged viral library DNA was isolated from the injected viral library by digestion of the viral capsid and purification of the contained ssDNA. These viral genomes were amplified by two PCR amplification steps, like the viral DNA extracted from tissue, to add adapters and indices for Illumina next-generation sequencing, and purified after gel electrophoresis. This viral library DNA, along with the viral DNA extracted from tissue, was sent for deep sequencing using an Illumina NextSeq 2000 system.

[0152] NGS data alignment and processing. Raw fastq files from NGS runs were processed with custom-built scripts (Capsida CapSeq Tools). For the first round library, the pipeline to process these datasets involved filtering to remove low-quality reads, utilizing a quality score for each sequence, and eliminating bias from PCR-induced mutations or high GC-content. The filtered dataset was then aligned by a perfect string match algorithm and trimmed to improve the alignment quality. Read counts for each sequence were pulled out and displayed by tissue, at which point all sequences found in the brain were compiled for formation of the second round library.

[0153] For the second round library read counts by tissue were similarly tabulated. Then, a read count of 1 was added to each sequence to remove 0 values, all brain regions for each sequence were summed together, and the read sequences for each codon replicate of a given 7-mer amino acid sequence were summed together to give a single value for each peptide insertion. Finally, the data was log 10 counts per million (Cpm) normalized. Enrichment values were calculated with normalized cpm of brain vs cpm of virus and converted to log 10.

[0154] While preferred embodiments of the present invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the invention. It should be understood that various alternatives to the embodiments of the invention described herein may be employed in practicing the invention. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby.

[0155] All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference.

Claims

1. An AAV capsid protein comprising a sequence as provided in Table 1 or FIG. 1.
2. The AAV capsid protein of claim 1, wherein the sequence is selected from TABLE-US-00006 (SEQ ID NO: 2) AQANTVRNTAQ, (SEQ ID NO: 3) AQDNHVRTVAQ, (SEQ ID NO: 4) AQENSTKSMAQ, (SEQ ID NO: 5) AQENTVRNMAQ, (SEQ ID NO: 6) AQGNAIRNGAQ, (SEQ ID NO: 7) AQGNAVKPFAQ, (SEQ ID NO: 8) AQGNGTRPTAQ, (SEQ ID NO: 9) AQHNNVRPVAQ, (SEQ ID NO: 10) AQLLRDKSSAQ, (SEQ ID NO: 11) AQMNNTAVLAQ, (SEQ ID NO: 12) AQNNTTKLAAQ, (SEQ ID NO: 13) AQNNTVKNLAAQ, (SEQ ID NO: 14) AQSNVIRNSAQ, (SEQ ID NO: 15) AQTLKSPATAQ, (SEQ ID NO: 16) AQTNSVKLIAQ, (SEQ ID NO: 15) AQTSVVPVGAQ, (SEQ ID NO: 18) AQYNSTKNMAQ, (SEQ ID NO: 19) AQANSVKNIAQ, and (SEQ ID NO: 20) AQLNNTTKPIDG.
3. An AAV capsid protein comprising an insertion sequence selected from ANTVRNT (SEQ ID NO: 21), DNHVRTV (SEQ ID NO: 22), ENSTKSM (SEQ ID NO: 23), ENTVRNM (SEQ ID NO: 24), GNAIRNG (SEQ ID NO: 25), GNAVKPF (SEQ ID NO: 26), GNGTRPT (SEQ ID NO: 27), HNNVRPV (SEQ ID NO: 28), LLRDKSS (SEQ ID NO: 29), MNNTAVL (SEQ ID NO: 30),

NNTTKLA (SEQ ID NO: 31), NNTVKNL (SEQ ID NO: 32), SNVIRNS (SEQ ID NO: 33), TLKSPAT (SEQ ID NO: 34), TNSVKLI (SEQ ID NO: 35), TSVVPVG (SEQ ID NO: 36), YNSTKNM (SEQ ID NO: 37), and ANSVKNI (SEQ ID NO: 38).

4. The AAV capsid protein of any of claims 1-3 wherein the AAV is AAV9.
5. The AAV capsid protein of any of claims 1-4 wherein the AAV is provided in SEQ ID NO: 1
6. The AAV capsid protein of any of claims 1-5 wherein the insertion is between amino acid 588 and amino acid 589.
7. The AAV capsid protein of any one of claims 1-6, wherein 60 copies of the AAV capsid protein are assembled into the AAV capsid.
8. The AAV capsid protein of any of claims 1-7, wherein the AAV capsid protein is present in VP1, VP2, and VP3 of the AAV capsid.
9. The AAV capsid protein of any of claims 1-8, further characterized by an increased transduction enrichment when measured in a CNS in a subject when delivered to the subject systemically.
10. The AAV capsid protein of any of claims 3-9, wherein the AAV capsid protein further comprises an amino acid substitution comprising A587D and/or Q588G.
11. An AAV capsid comprising an AAV capsid protein of any of claims 1-10.
12. The AAV capsid of claim 11 wherein the AAV capsid is chimeric.
13. The AAV capsid of any of claims 11-12 that is isolated and purified.
14. The AAV capsid of any of claims 11-13 formulated as a pharmaceutical formulation for systemic administration to treat a disease or a condition of the CNS, the pharmaceutical formulation further comprising a pharmaceutically acceptable carrier.
15. The AAV capsid protein of any one of claims 1-10, wherein the capsid protein is expressed in the CNS.
16. The AAV capsid protein of claim 15, wherein the CNS comprises a cell-type selected from a neuron, an oligodendrocyte, an astrocyte, and a brain vascular cell.
17. The AAV capsid protein of any one of claims 15-16, wherein the CNS comprises a tissue that is selected from brain, thalamus, cortex, putamen, lateral ventricles, medulla, the pons, the amygdala, the motor cortex, caudate, hypothalamus, striatum, ventral midbrain, neocortex, basal ganglia, hippocampus, thalamus, cerebrum, cerebellum, brain stem, and spinal cord.
18. A nucleic acid sequence encoding the peptide of any of claims 1-10, selected from the group consisting of SEQ ID NOs 67-85.
19. A nucleic acid sequence encoding a peptide sequence as provided in Table 1 or FIG. 1.
20. A recombinant vector comprising a nucleic acid encoding the AAV capsid protein of any one of claims 1-10.
21. A kit comprising: a) a first vector comprising the recombinant vector of claim 20; b) a second vector encoding a helper virus protein; and c) a third vector comprising a therapeutic nucleic acid encoding a therapeutic gene expression product.
22. A method of treating a disease or condition in a subject comprising administering a therapeutically effective amount of a pharmaceutical formulation comprising the AAV capsid protein of any one of claims 1-10.
23. The method of claim 22, wherein the disease or the condition is a disease or a condition of a CNS of the subject.
24. A method of manufacturing a recombinant AAV particle from the AAV capsid of any one of claims 11-13, the method comprising: a. introducing into a cell a nucleic acid comprising: i. a first nucleic acid sequence encoding a therapeutic gene expression product; ii. a second nucleic acid sequence encoding a recombinant viral genome comprising a capsid (Cap) gene modified to express the AAV capsid of any one of claims 11-12; and iii. a third nucleic acid sequence encoding an AAV helper virus genome; and b. assembling the recombinant AAV particle, the recombinant AAV particle comprising the AAV capsid encapsidating the first nucleic acid.
25. The AAV capsid protein of any of claims 1-10 and 15-17 wherein the AAV capsid protein is

characterized by an increased transduction enrichment when measured in a tissue in a subject when delivered to the subject systemically.

26. The AAV capsid protein of any of claims 1-10 and 15-17 wherein the tissue is brain tissue.

27. An AAV particle comprising an AAV capsid protein of any of claims 1-10 and 15-17 and a viral genome.

28. A peptide comprising an amino acid sequence selected from ANTVRNT (SEQ ID NO: 21), DNHVRTV (SEQ ID NO: 22), ENSTKSM (SEQ ID NO: 23), ENTVRNM (SEQ ID NO: 24), GNAIRNG (SEQ ID NO: 25), GNAVVPF (SEQ ID NO: 26), GNGTRPT (SEQ ID NO: 27), HNNVRPV (SEQ ID NO: 28), LLRDKSS (SEQ ID NO: 29), MNNTAVL (SEQ ID NO: 30), NNTTKLA (SEQ ID NO: 31), NNTVKNL (SEQ ID NO: 32), SNVIRNS (SEQ ID NO: 33), TLKSPAT (SEQ ID NO: 34), TNSVKLI (SEQ ID NO: 35), TSVVPVG (SEQ ID NO: 36), YNSTKNM (SEQ ID NO: 37), and ANSVKNI (SEQ ID NO: 38).

29. A recombinant adeno-associated virus (rAAV) comprising a capsid comprising any one of TABLE-US-00007 (SEQ ID NO: 21) ANTVRNT, (SEQ ID NO: 22) DNHVRTV, (SEQ ID NO: 23) ENSTKSM, (SEQ ID NO: 24) ENTVRNM, (SEQ ID NO: 25) GNAIRNG, (SEQ ID NO: 26) GNAVVPF, (SEQ ID NO: 27) GNGTRPT, (SEQ ID NO: 28) HNNVRPV, (SEQ ID NO: 29) LLRDKSS, (SEQ ID NO: 30) MNNTAVL, (SEQ ID NO: 31) NNTTKLA, (SEQ ID NO: 32) NNTVKNL, (SEQ ID NO: 33) SNVIRNS, (SEQ ID NO: 34) TLKSPA, (SEQ ID NO: 35) TNSVKLI, (SEQ ID NO: 36) TSVVPVG, (SEQ ID NO: 37) YNSTKNM, and (SEQ ID NO: 38) ANSVKNI.

30. A recombinant adeno-associated virus (rAAV) comprising a capsid comprising any one of TABLE-US-00008 (SEQ ID NO: 2) AQANTVRNTAQ, (SEQ ID NO: 3) AQDNHVRTVAQ, (SEQ ID NO: 4) AQENSTKSMAQ, (SEQ ID NO: 5) AQENTVRNMAQ, (SEQ ID NO: 6) AQGNAIRNGAQ, (SEQ ID NO: 7) AQGNAVVPFAQ, (SEQ ID NO: 8) AQGNGTRPTAQ, (SEQ ID NO: 9) AQHNNVRPVAQ, (SEQ ID NO: 10) AQLLRDKSSAQ, (SEQ ID NO: 11) AQMNNTAVLAQ, (SEQ ID NO: 12) AQNNTTKLAAQ, (SEQ ID NO: 13) AQNNTVKNLAQ, (SEQ ID NO: 14) AQSNVIRNSAQ, (SEQ ID NO: 15) AQTLKSPATAQ, (SEQ ID NO: 16) AQTNSVKLIAQ, (SEQ ID NO: 15) AQTSVVPVGAQ, (SEQ ID NO: 18) AQYNSTKNMAQ, (SEQ ID NO: 19) AQANSVKNIQAQ, and (SEQ ID NO: 20) AQLNNTTKPIDG.
